

Anatomy of the Male & Female Perineum



Wessam Ibrahim, MD, PhD, MEHP

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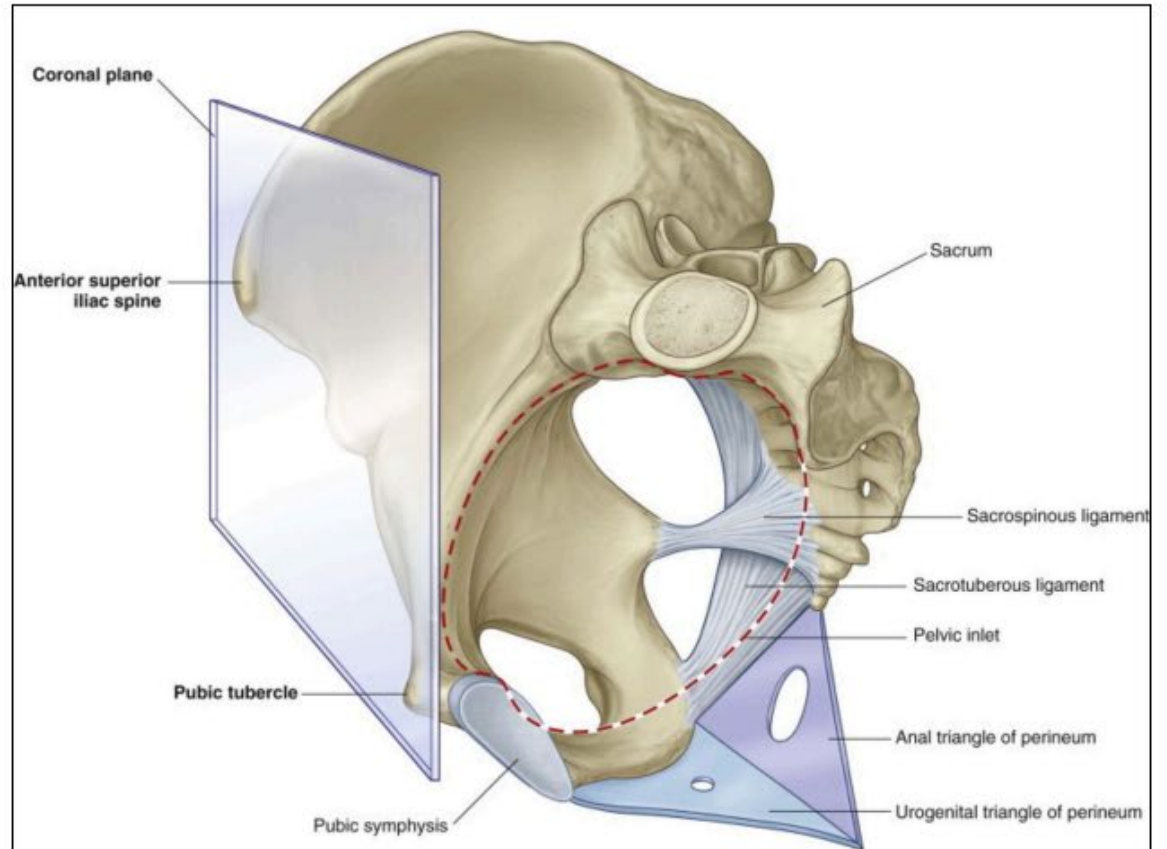
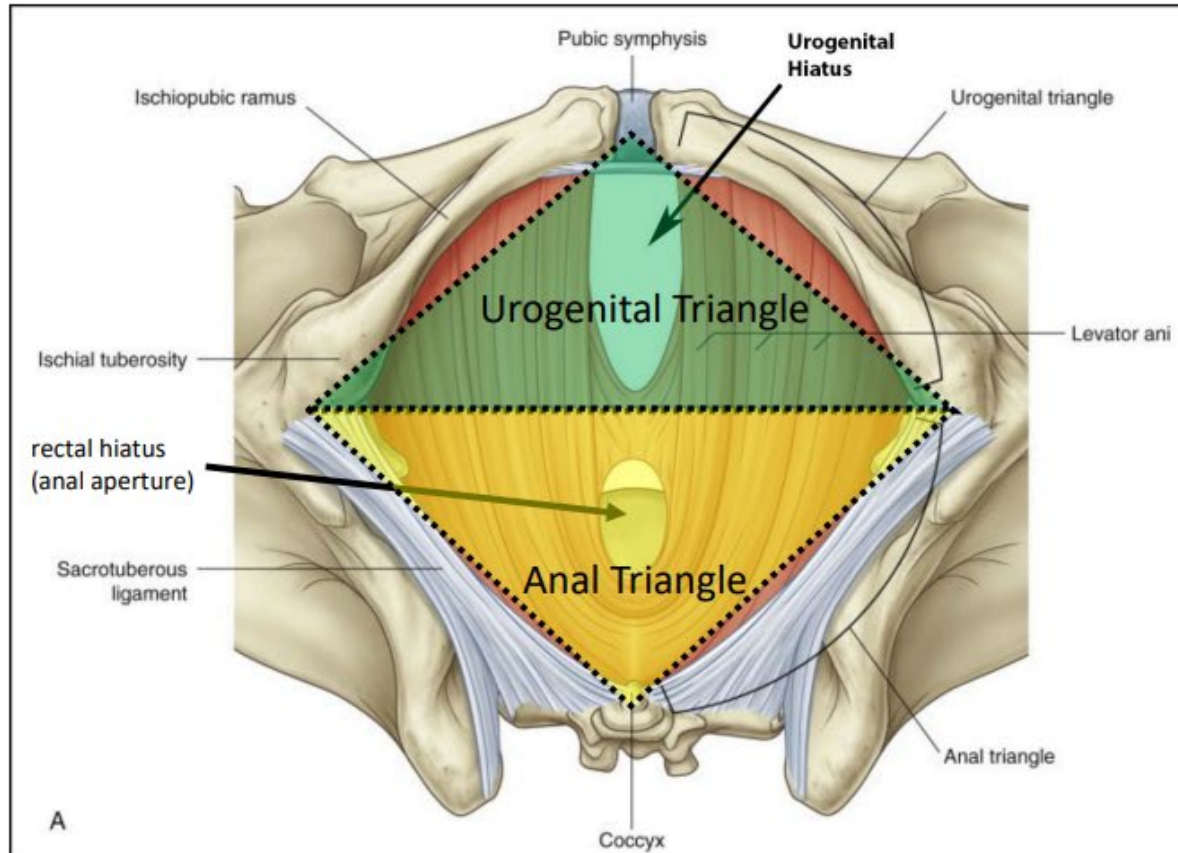
Learning Objectives

- Differentiate the subdivisions of the perineum (urogenital and anal regions) based on their boundaries and spatial orientation.
- Describe the attachment of the perineal membrane.
- Compare the blood supply, venous drainage, and lymphatic drainage of the anal canal with particular attention to the differences superior and inferior to the pectinate line.
- Describe the anatomy of female external genitalia (mons pubis, labia majora, labia minora, vestibule, clitoris, urethra, bulbs of the vestibule, greater vestibular (Bartholin's) glands, vagina).
- Describe the anatomy of male external genitalia (penis, testicles, epididymis, and course of the male urethra).
- Explain the innervation, blood supply, and venous drainage of the external genitalia, emphasizing the role of the pudendal nerves and vessels.
- Trace the lymphatic drainage of pelvic and perineal organs in males and females, including the clinical applications of affected lymph nodes in various invasive cancers and infections of internal pelvic structures.
- Describe the autonomic innervation of the pelvic & perineal regions and the effects of sympathetic vs. parasympathetic input on pelvic organs.

Perineum

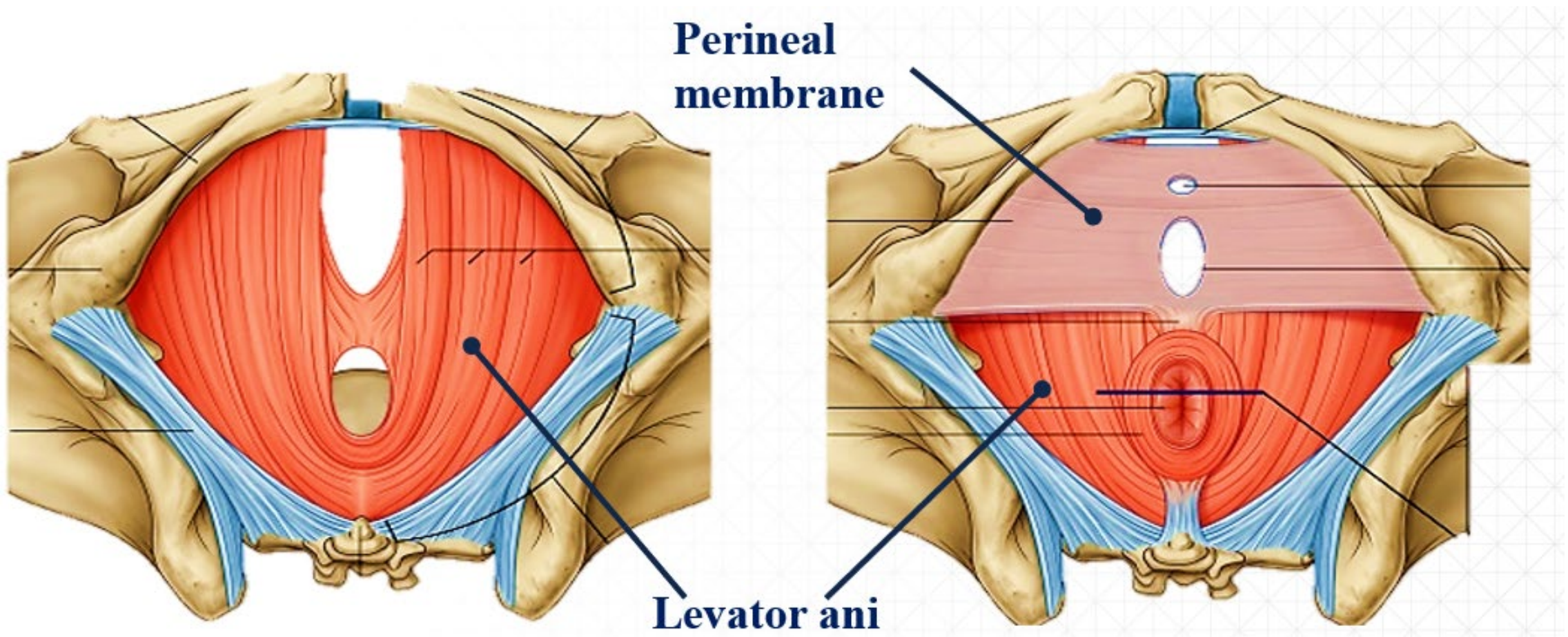
The urogenital triangle is the anterior triangular region of the diamond-shaped perineum. In anatomical position, it is oriented in a horizontal plane & contains the urethra and external genitalia.

The anal triangle is the posterior triangular region. In anatomical position, It is oriented in a plane that is almost vertical and contains the anal canal, anus, external anal sphincter, inferior rectal nerve (branch of pudendal nerve), and the fat-filled ischiorectal fossa.

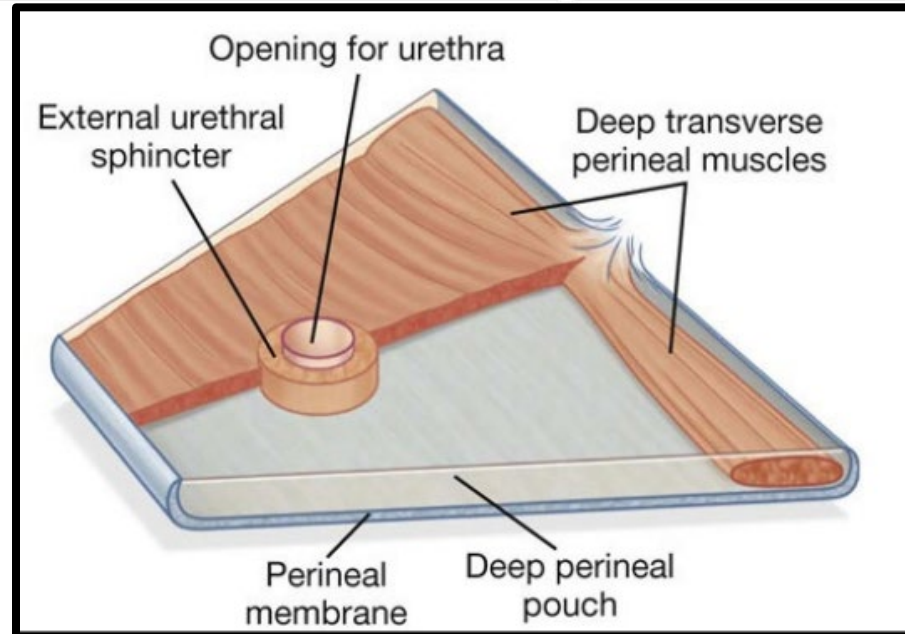
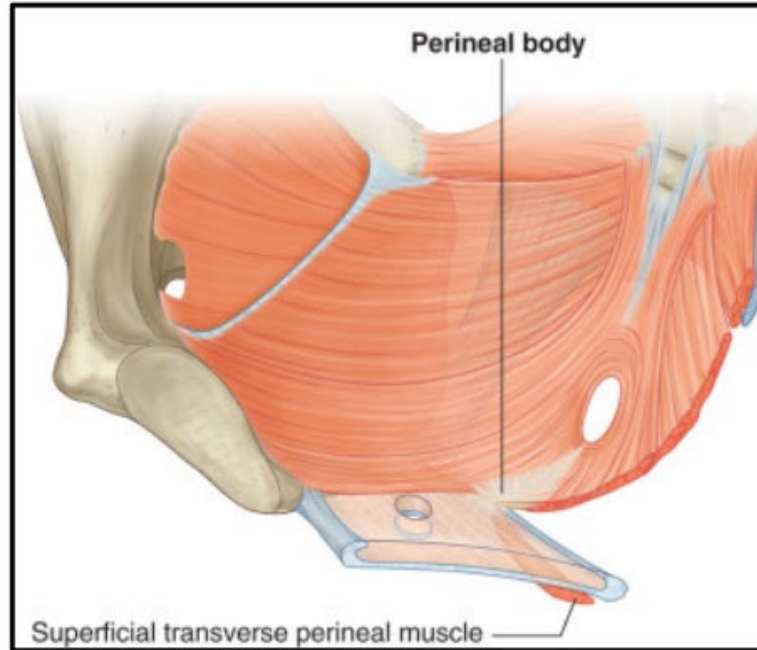
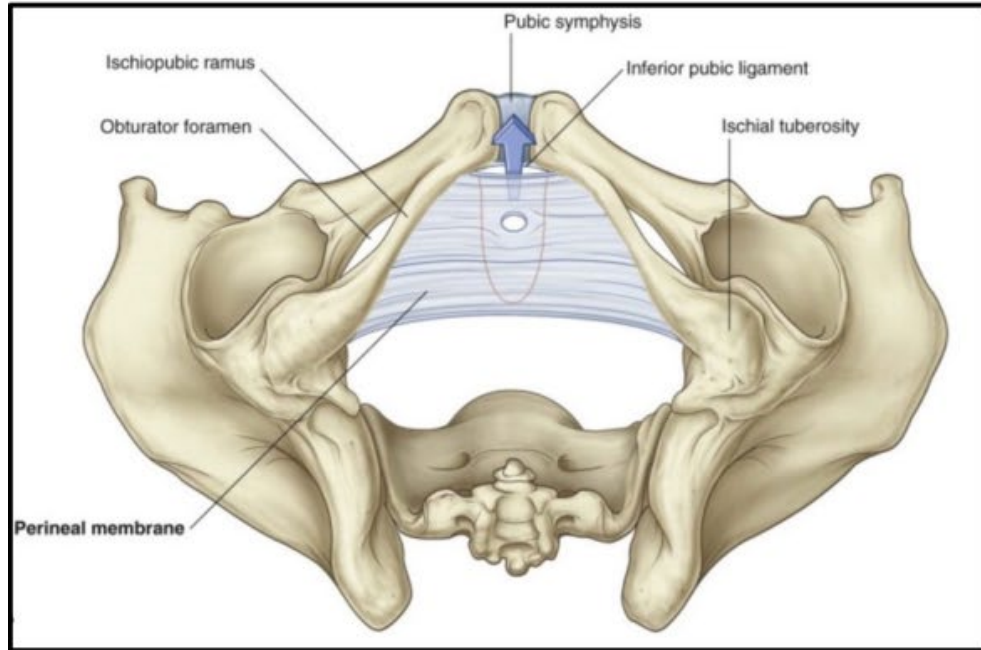


Urogenital Triangle: Perineal Membrane

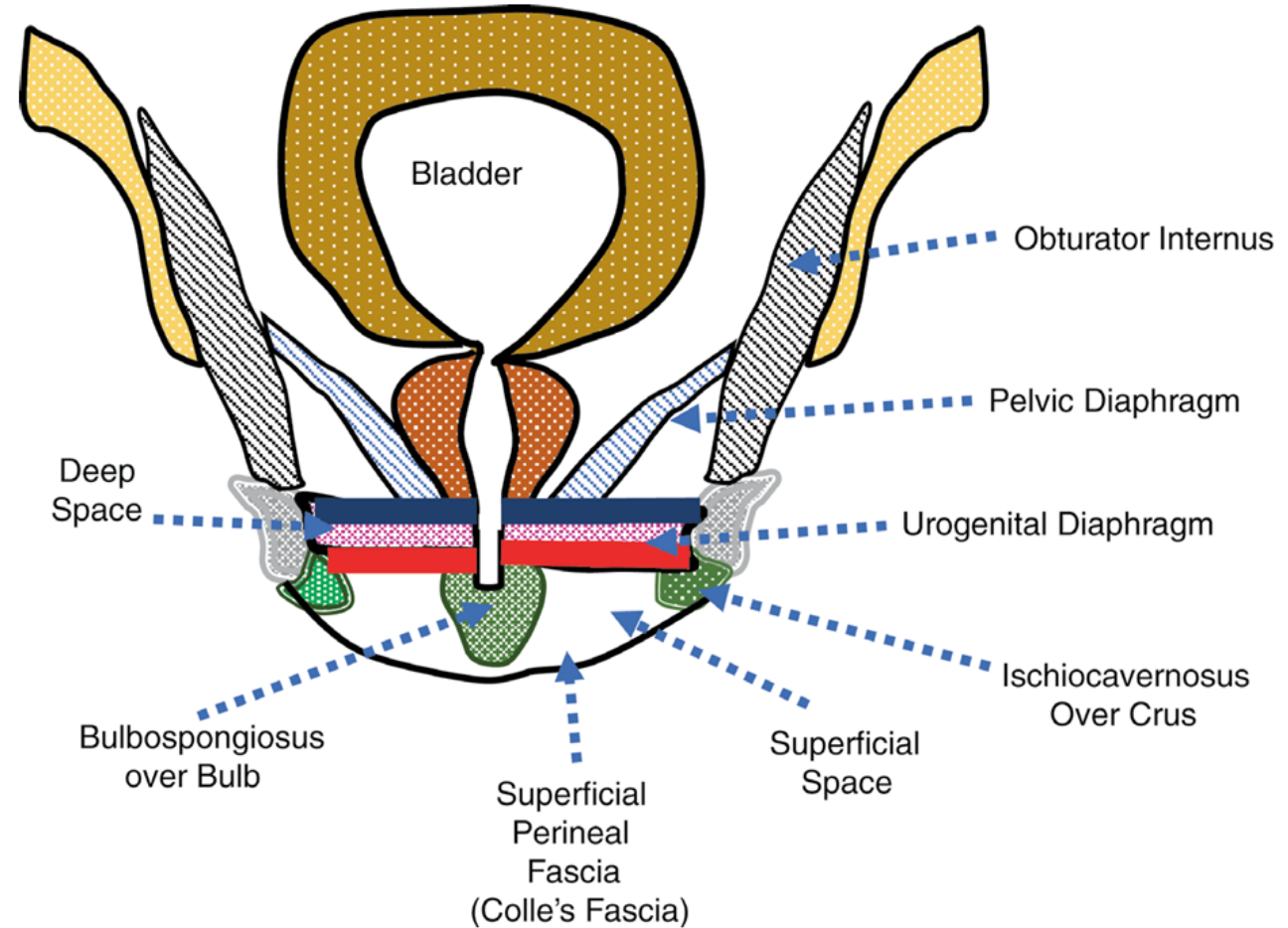
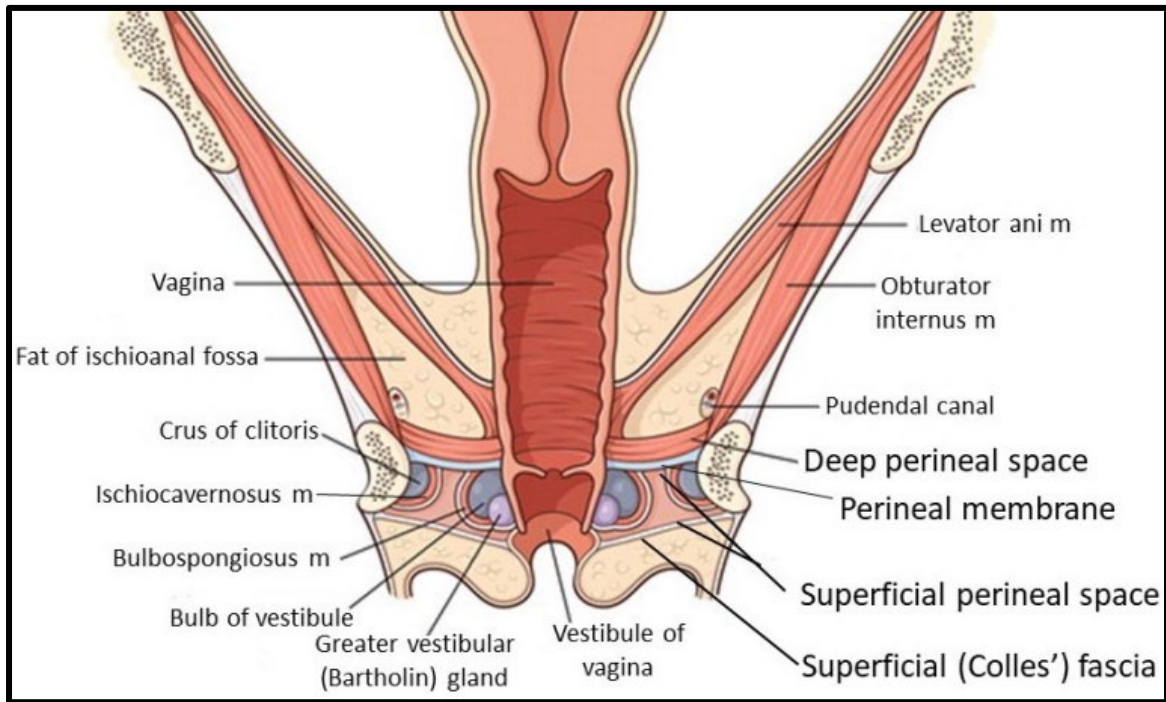
Perineal membrane: thin fibromuscular horizontal layer with fibers running parallel to pelvic diaphragm (i.e., levator ani), divides urogenital triangle into a deep (superior to membrane) & superficial (inferior to membrane) pouch

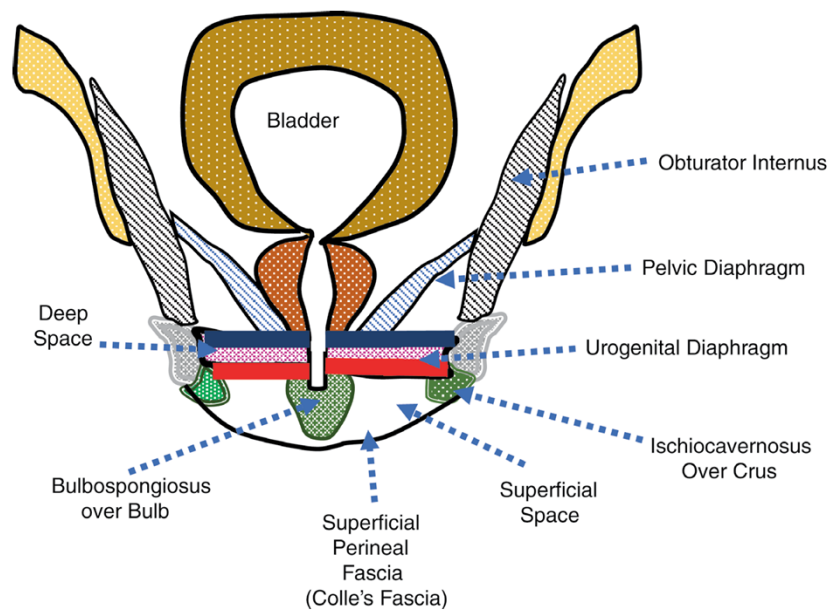
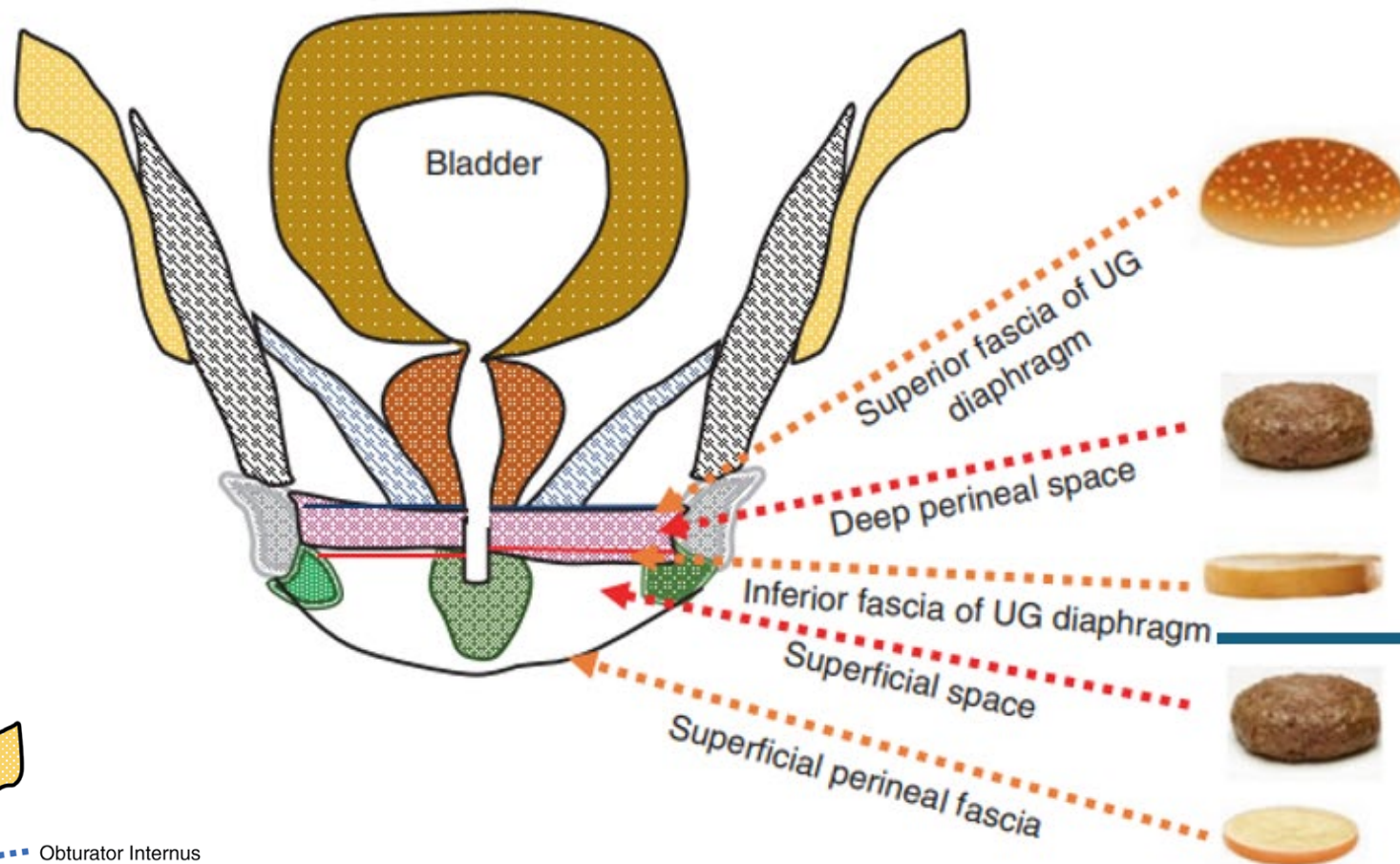


Perineum Overview



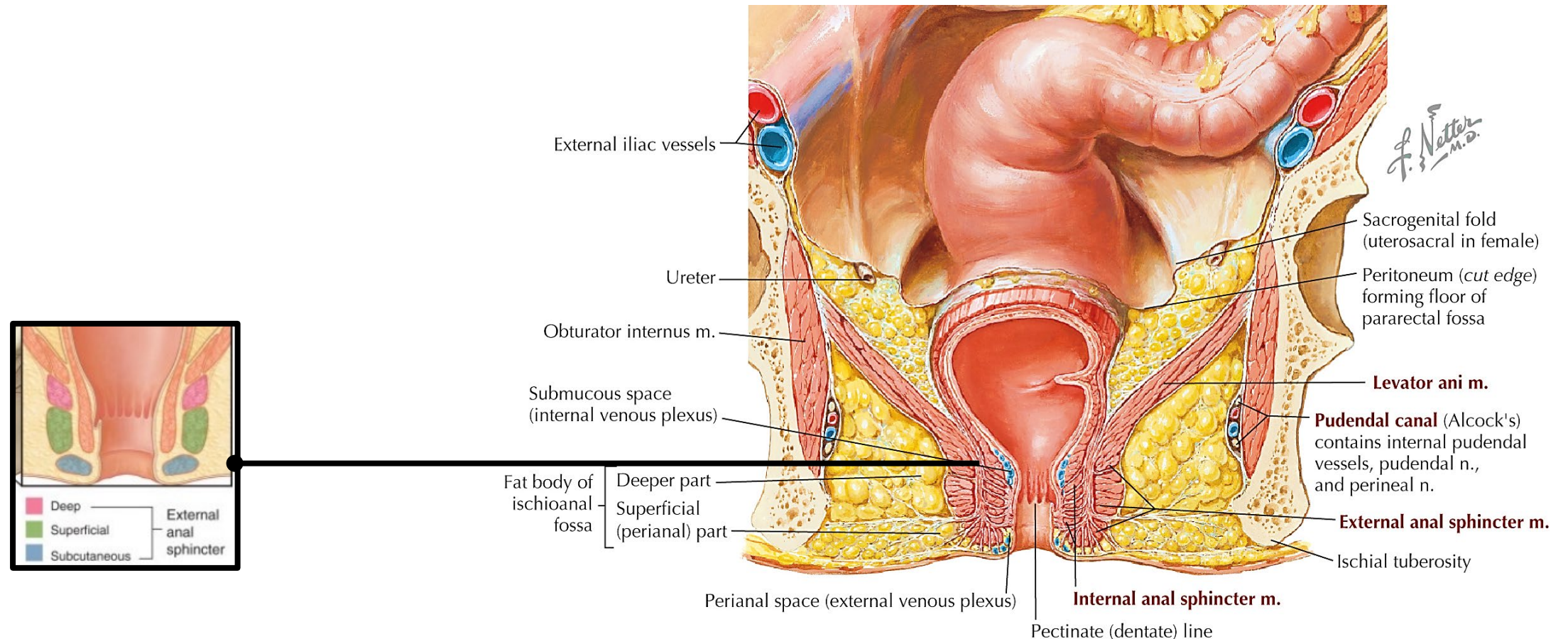
Perineum Overview



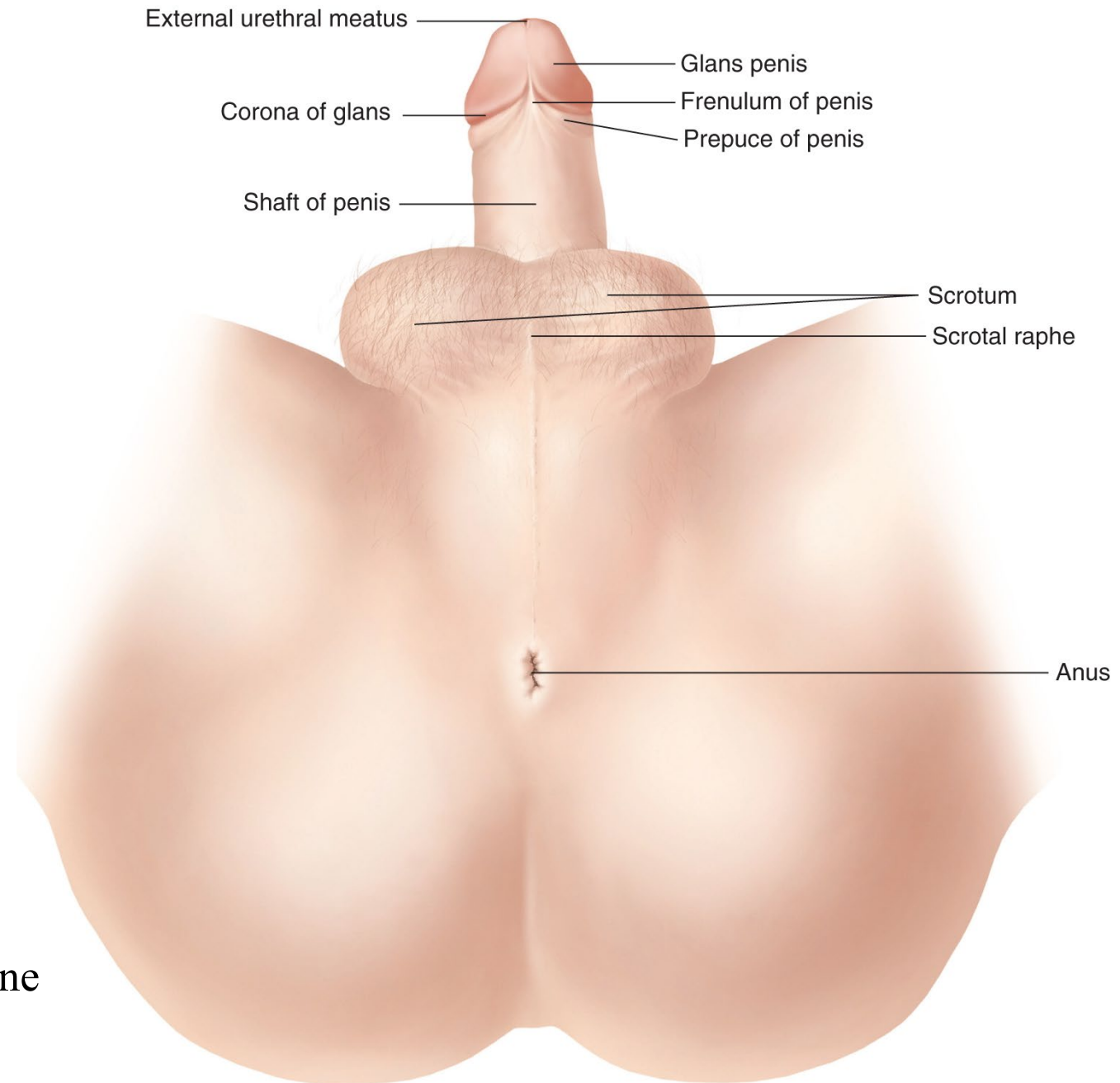
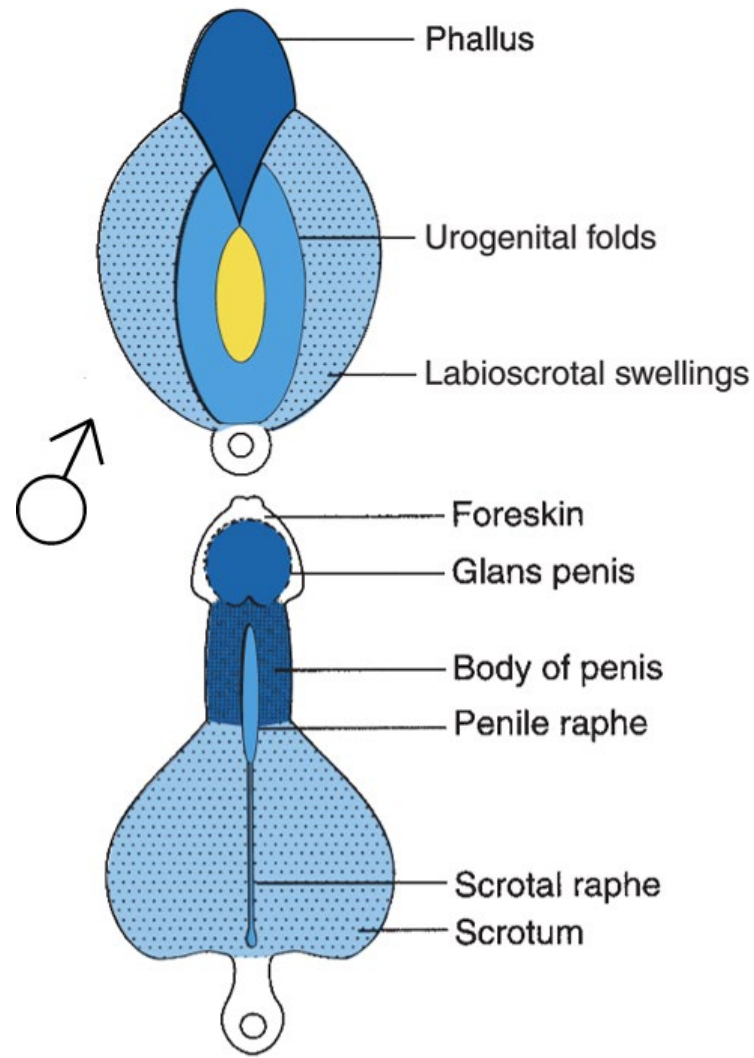


Ischioanal (Ischiorectal) Fossa

- Fossa filled with fat, poorly vascularized.
- Allows for expansion of the anal canal during defecation & accommodates the fetus during childbirth.
- Infections from one of the fossae can spread to the other & into the urogenital triangle.
- **Contents:** Fat, pudendal nerves and vessels giving the inferior rectal nerve & vessels



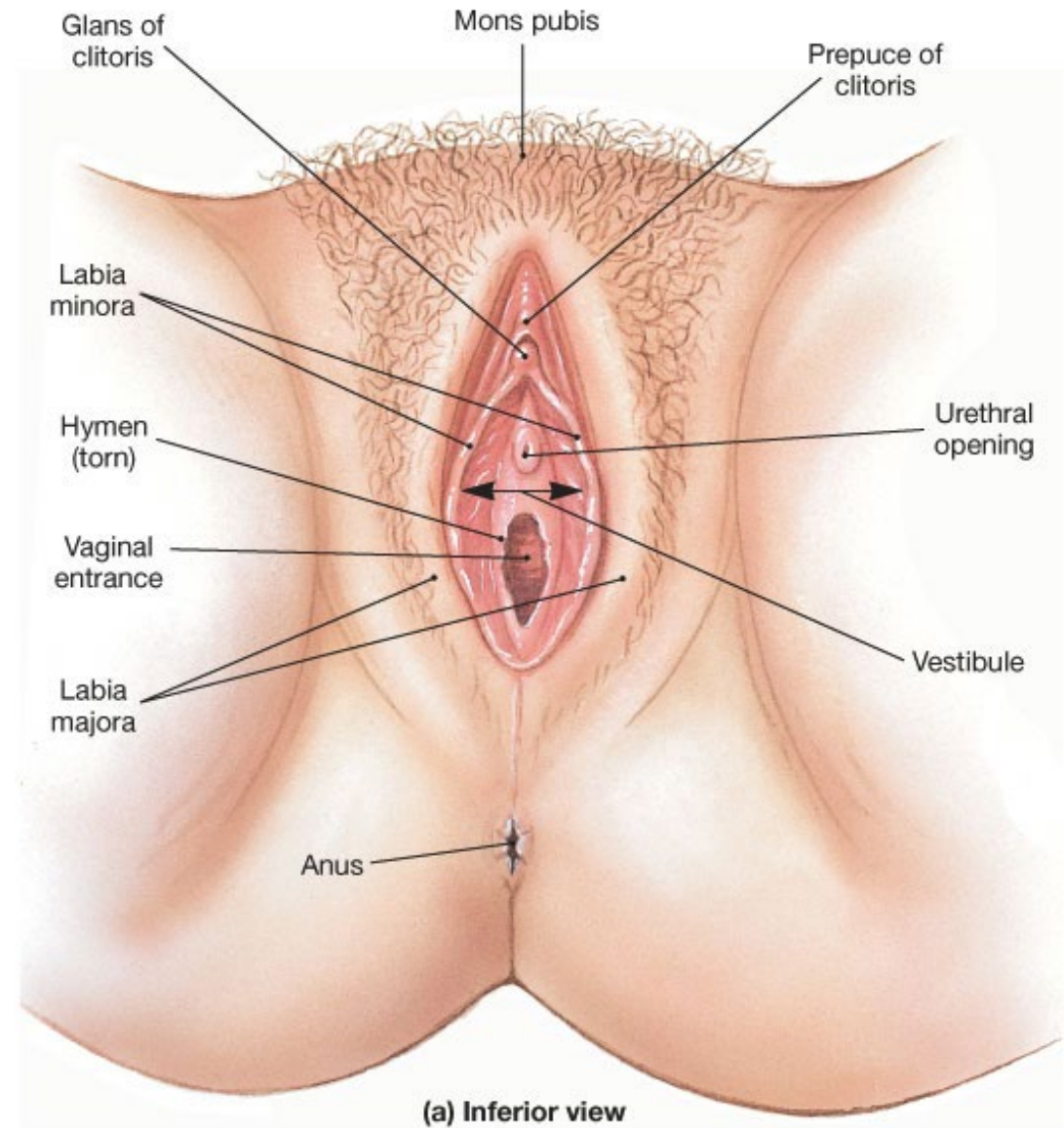
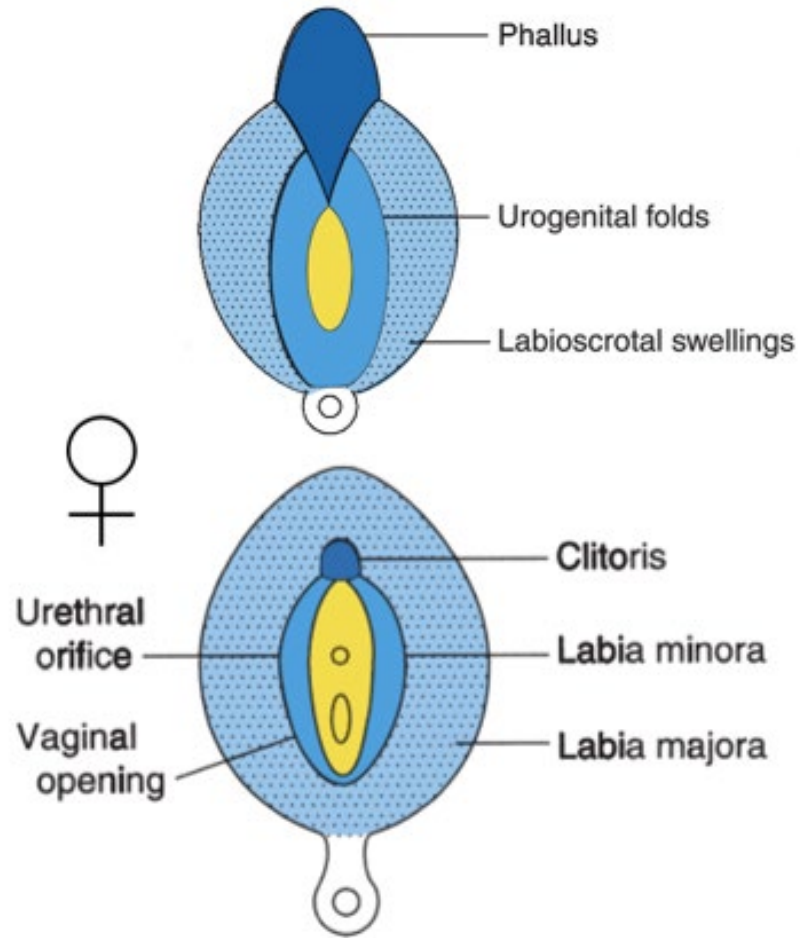
Male Surface Anatomy



Proliferation of mesoderm around the cloacal membrane causes the overlying ectoderm to form three elevated structures. Stimulated by testosterone



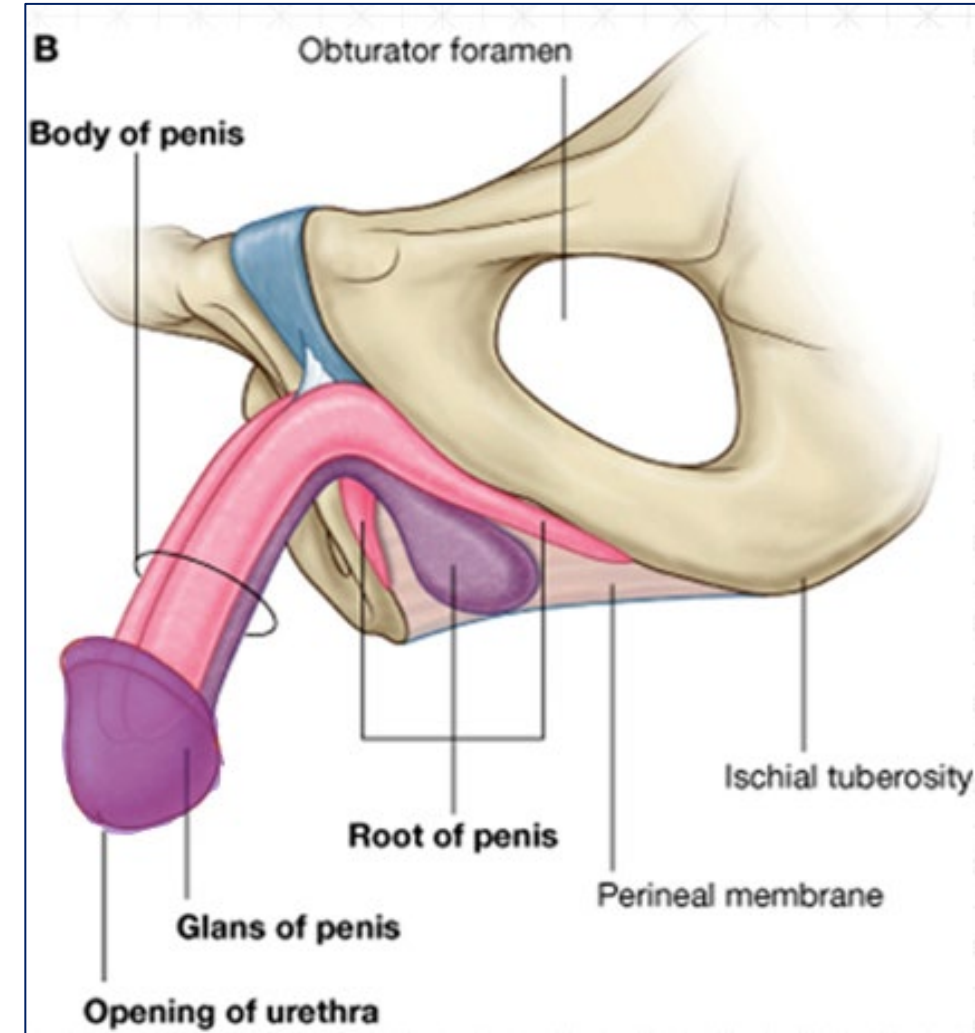
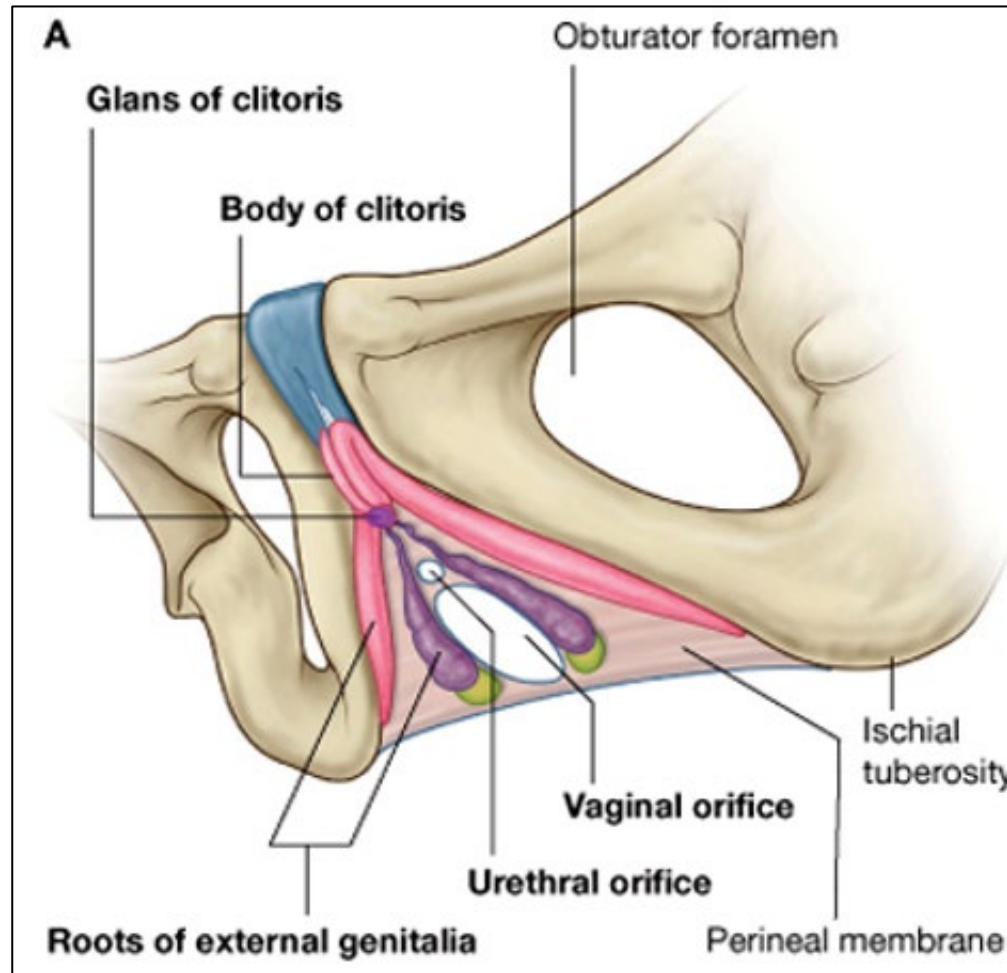
Female Surface Anatomy



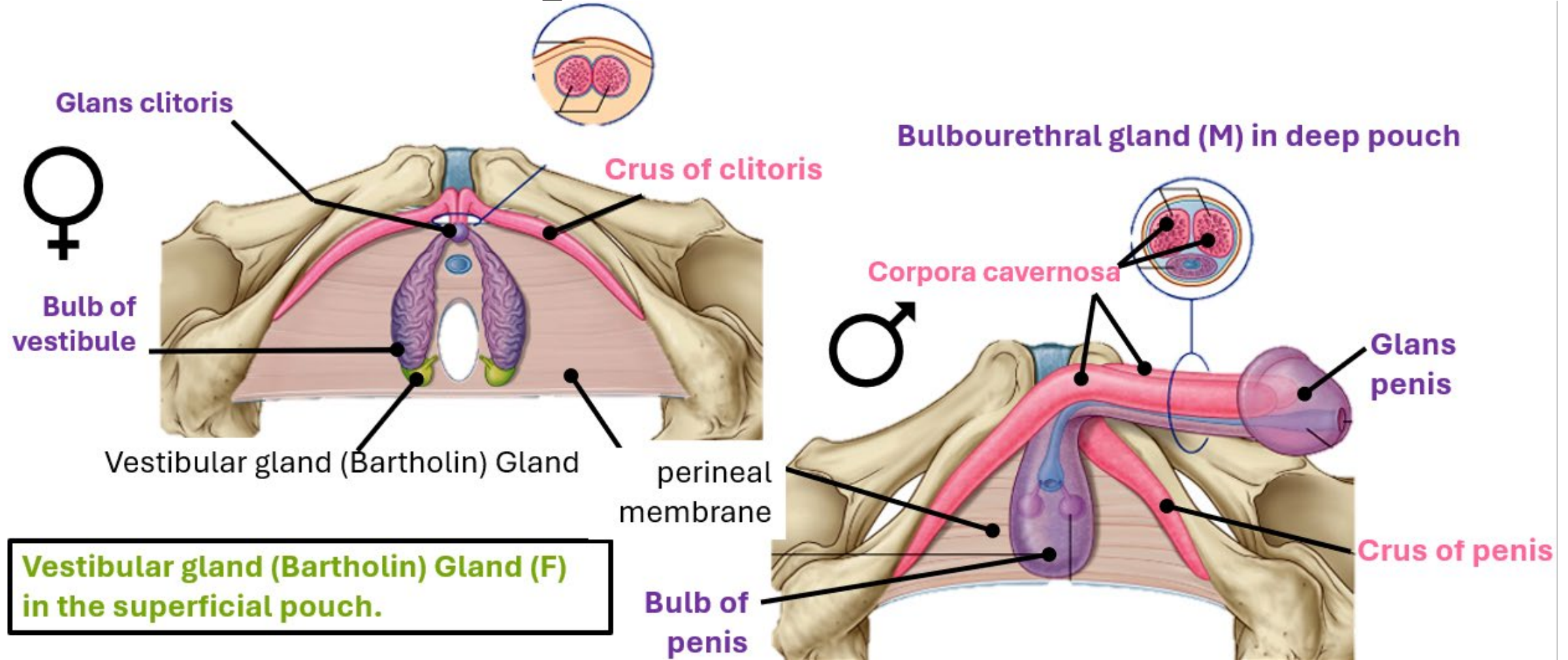
Female & Male Superficial Pouch

Male: **Bulb** & **crura** (attached) of the penis give rise to erectile sinuses of the penis – **corpus spongiosum** & **corpora cavernosa** (free), respectively.

Female: **Bulb** of the vestibule & **crura** (attached) give rise to erectile sinuses of clitoris – **glans clitoris** & **corpora cavernosa** (free) respectively.



Female & Male Superficial Pouch: Erectile Tissue



Root/attached components: sit/attach to the perineal membrane & are the bulb and crus.

Free components: **corpora cavernosa** - body of the clitoris (F) & body of the penis (M), and **corpus spongiosum** –glans clitoris (F) and glans penis (M).

Dorsal vein of
penis, superficial

Dorsal vein of
penis, deep

Corpus
cavernosum

Deep artery of
penis

Corpus
spongiosum

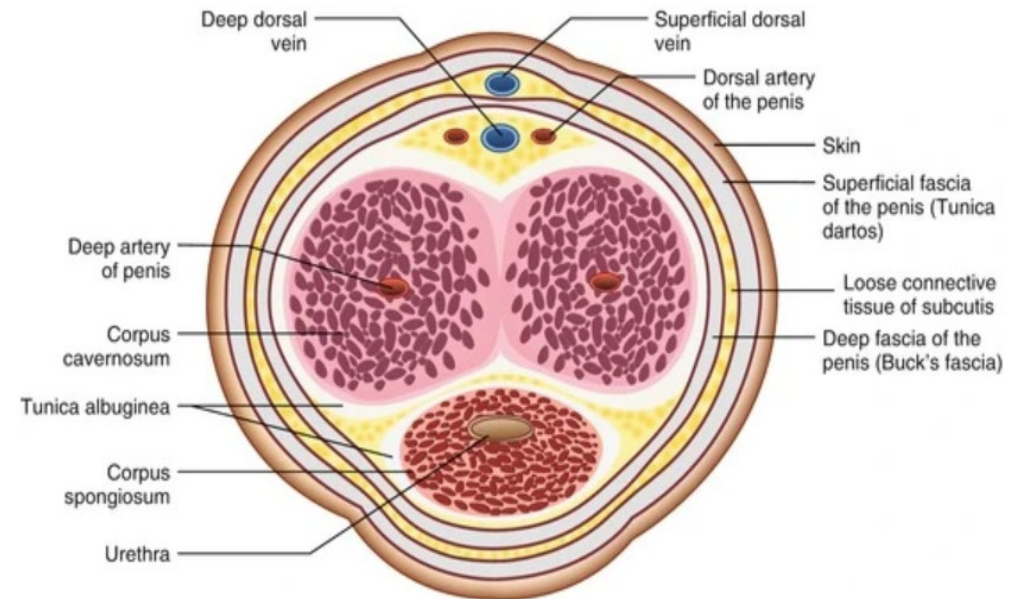
Dorsal arteries
of penis

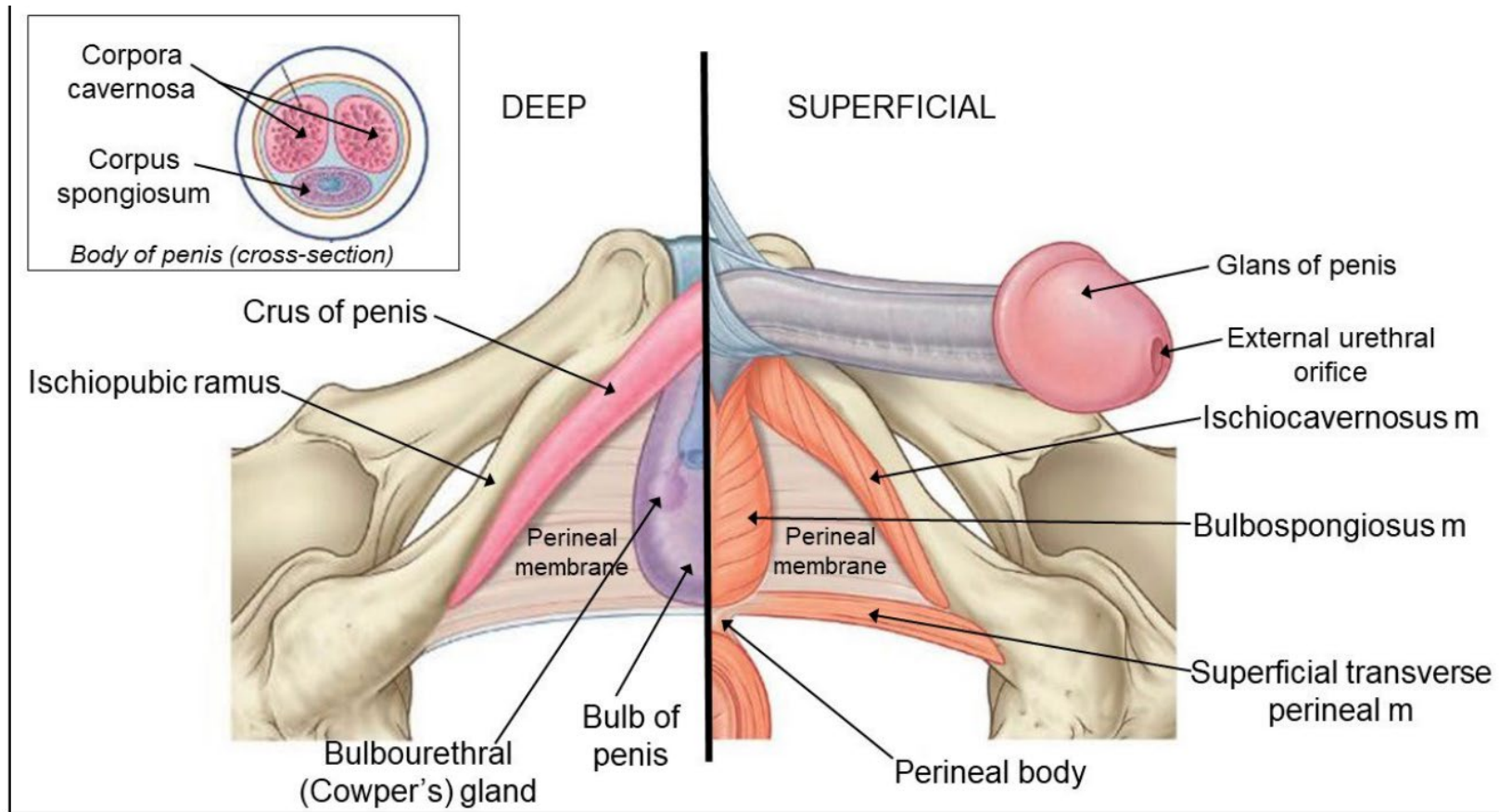
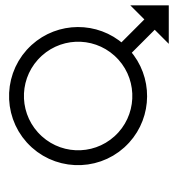
Urethra,
spongy part

Transverse section of penis

BlueLink

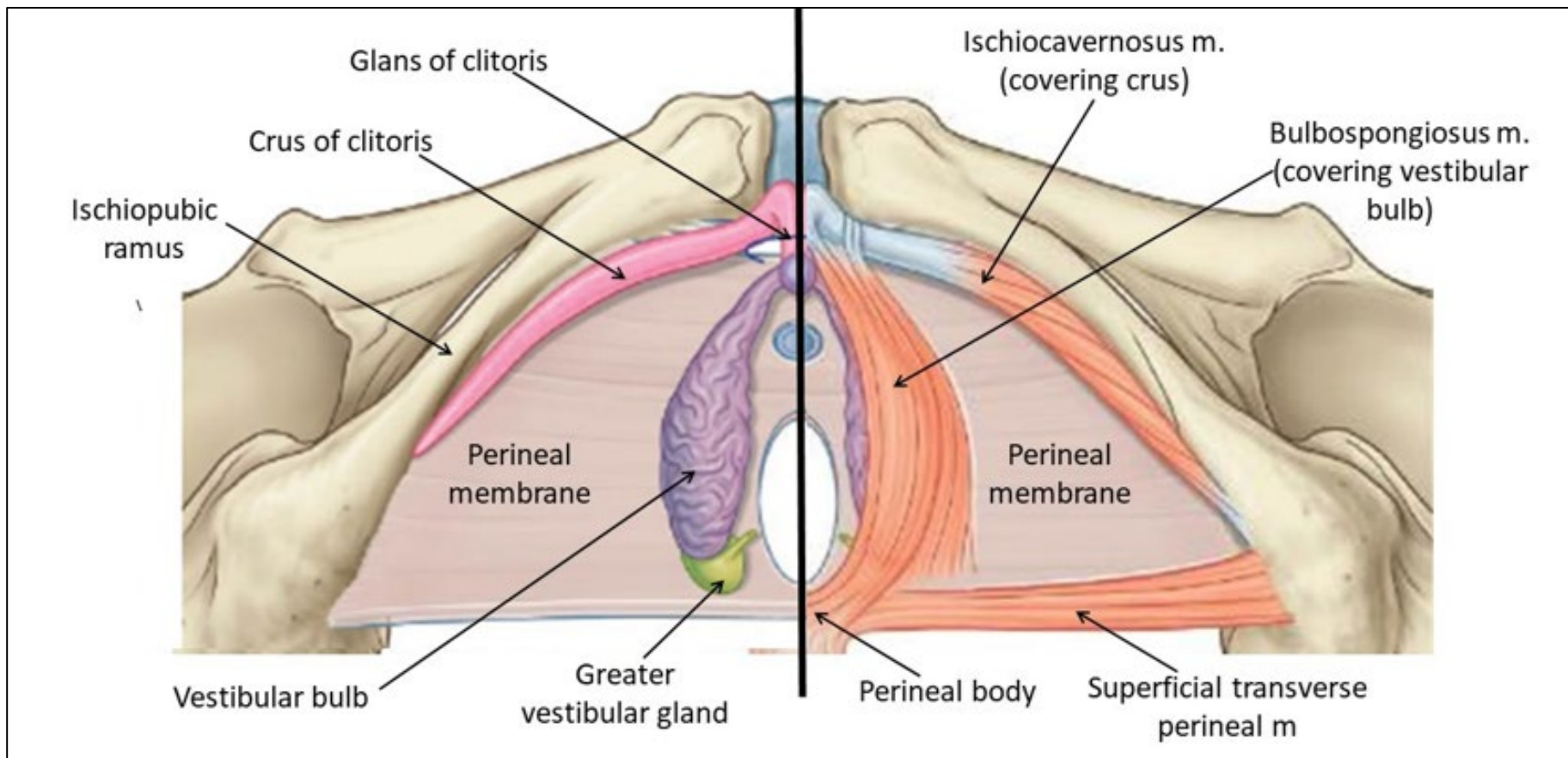
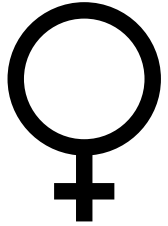
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Superficial pouch contains external genitalia & associated skeletal muscles: **bulbospongiosus**, **ischiocavernosus**, & **superficial transverse perineal muscles**. Because of their action during urination/erection, these muscles are generally more developed in males.

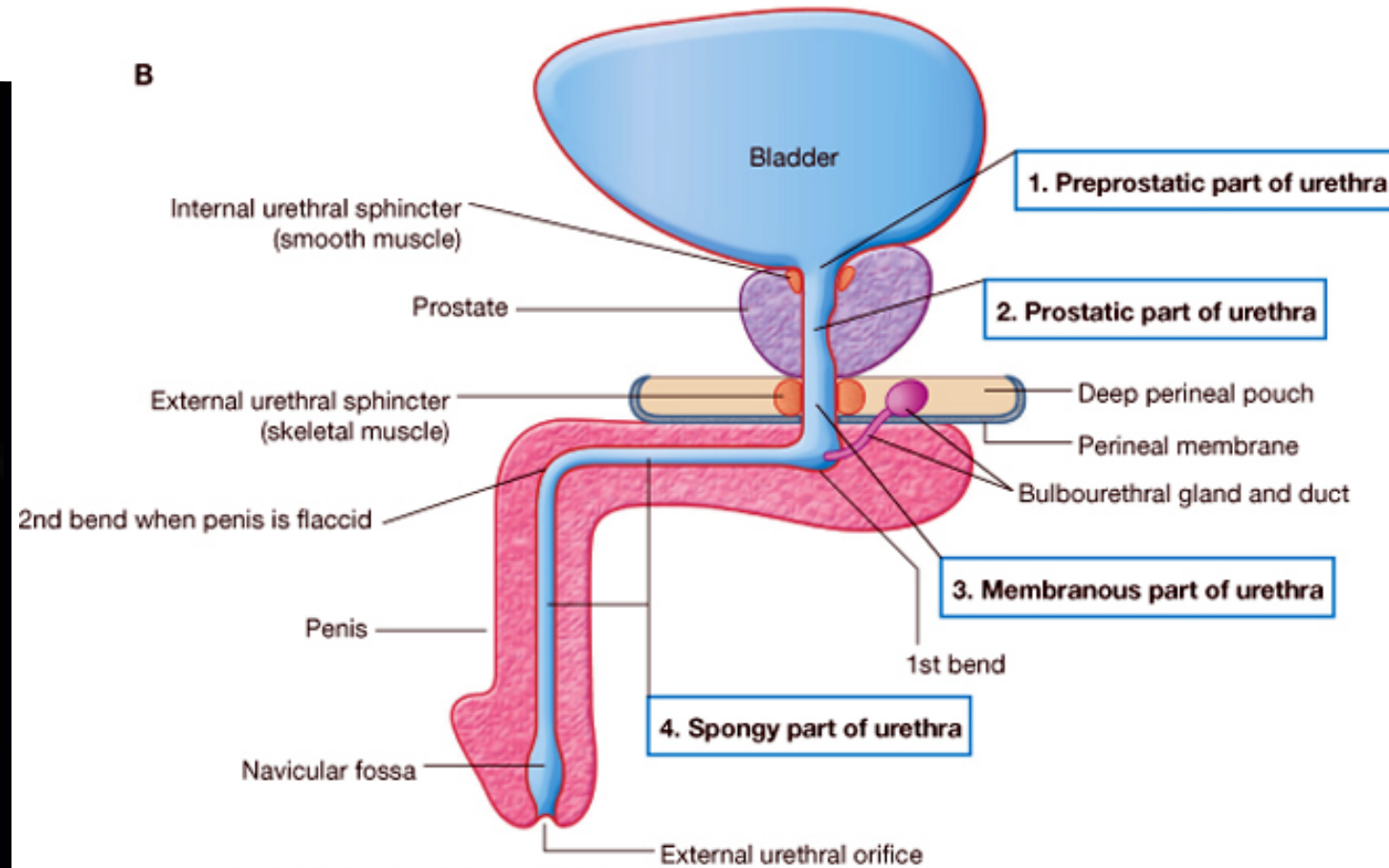
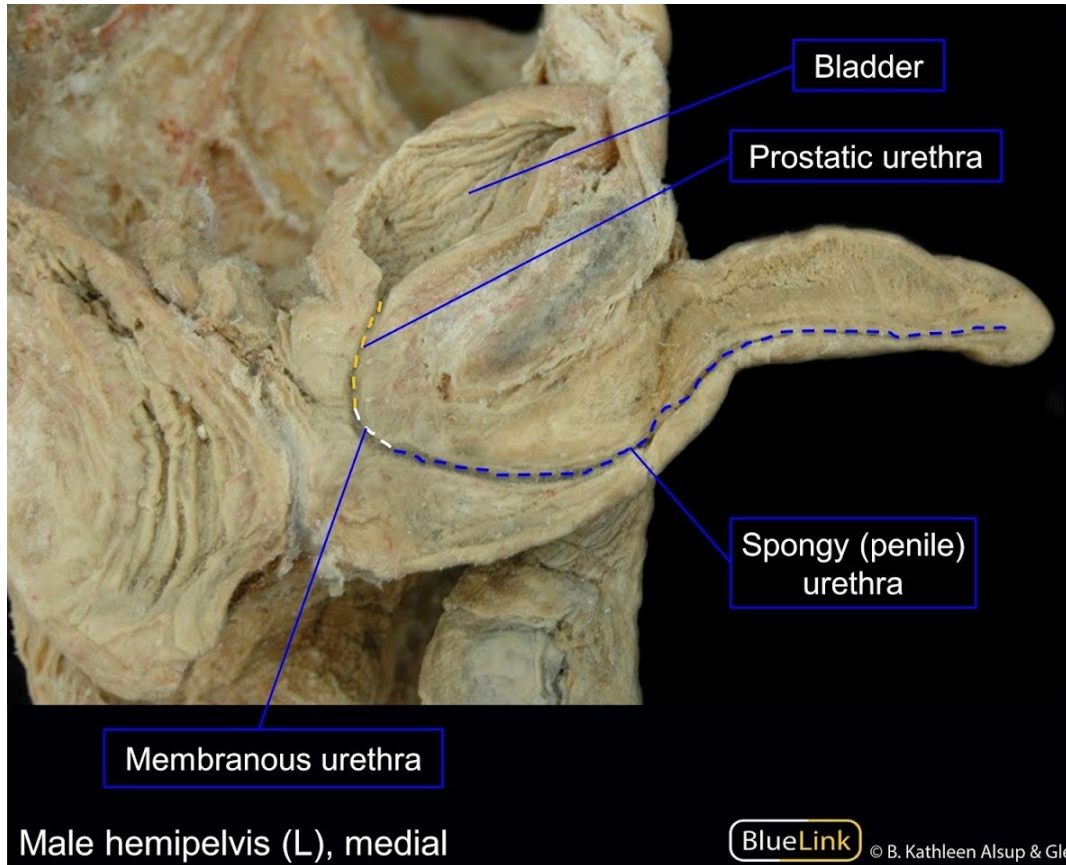
- **Sup. transverse perineal muscles:** provide a firm base for the penis during erection
- **Bulbospongiosus:** aids in emptying of urine/semen.
- **Ishiocavernosus:** force blood from crura to corpora cavernosa, increasing turgidity of penis.



Superficial pouch contains external genitalia & associated skeletal muscles, including **bulbospongiosus, ischiocavernosus, & superficial transverse perineal muscles**. Muscles cover erectile tissue.

- **Sup. transverse perineal muscle:** support pelvic floor, fix perineal body
- **Bulbospongiosus:** “sphincter” of vagina aids in erection, compresses greater vestibular gland.
- **Ishiocavernosus:** forces blood from crura to corpora cavernosa, maintains erection of clitoris.

Male Urethra



Drake: Gray's Anatomy for Students, 2nd Edition.
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Male & Female Deep Pouch

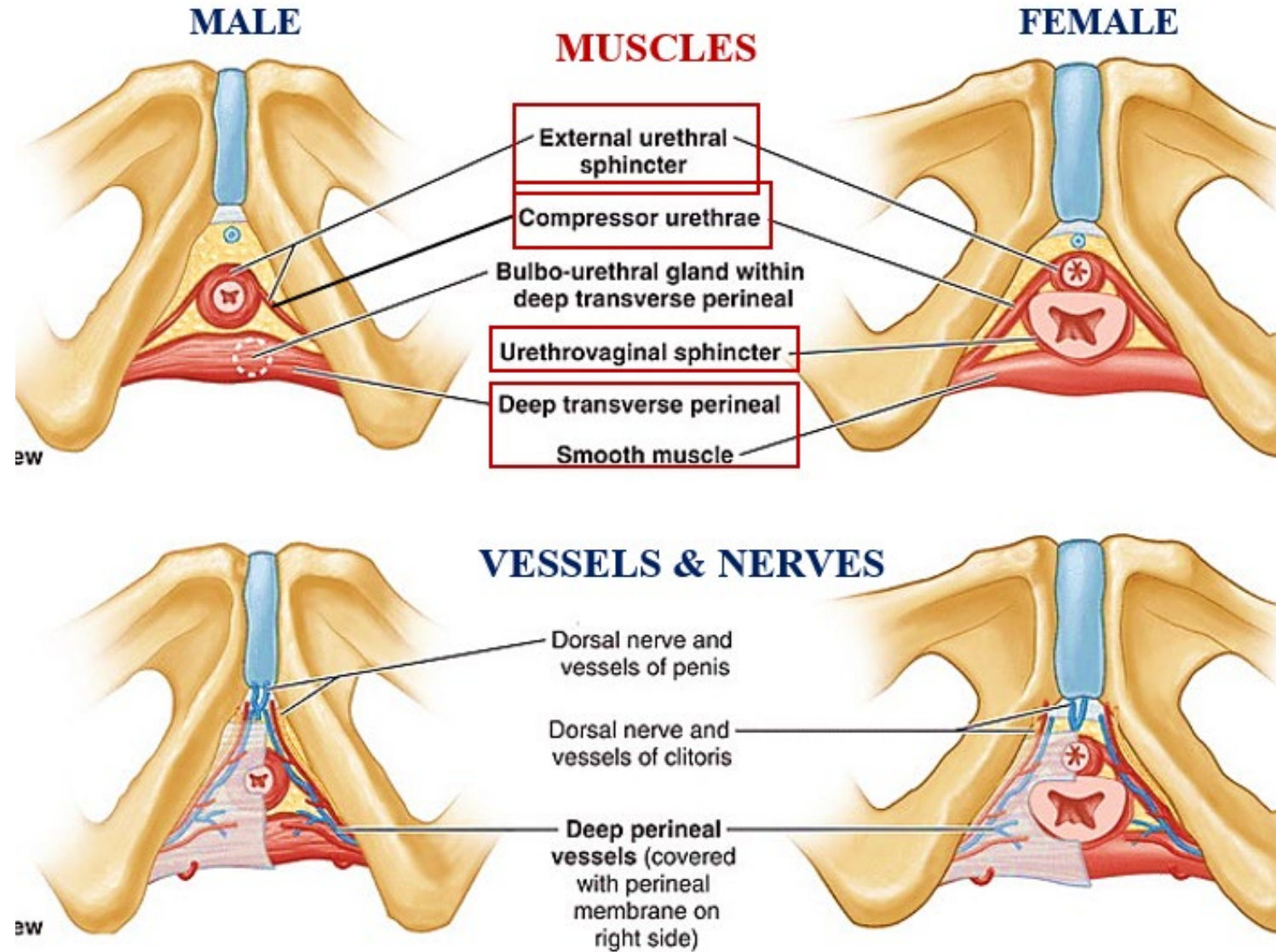
- Fat of the ischioanal fossae
- Dorsal vessels of penis/clitoris from the internal pudendal a.
- Dorsal nerve of penis/clitoris from the pudendal nerve
- Deep transverse perineal m
- Sphincter urethrae



- Membranous urethra
- Bulbourethral glands

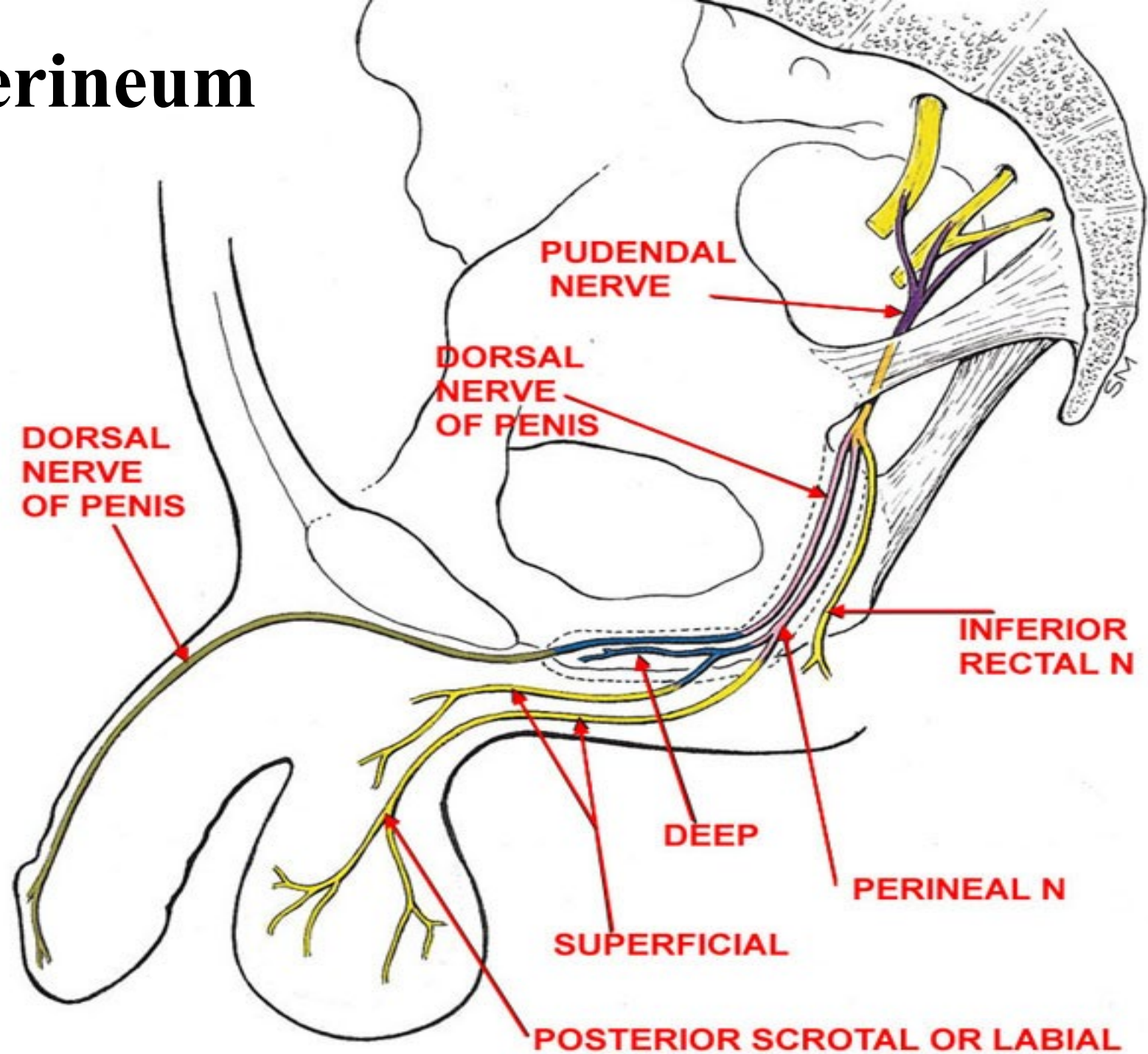


- openings for urethra & vagina
- Urethrovaginal sphincter

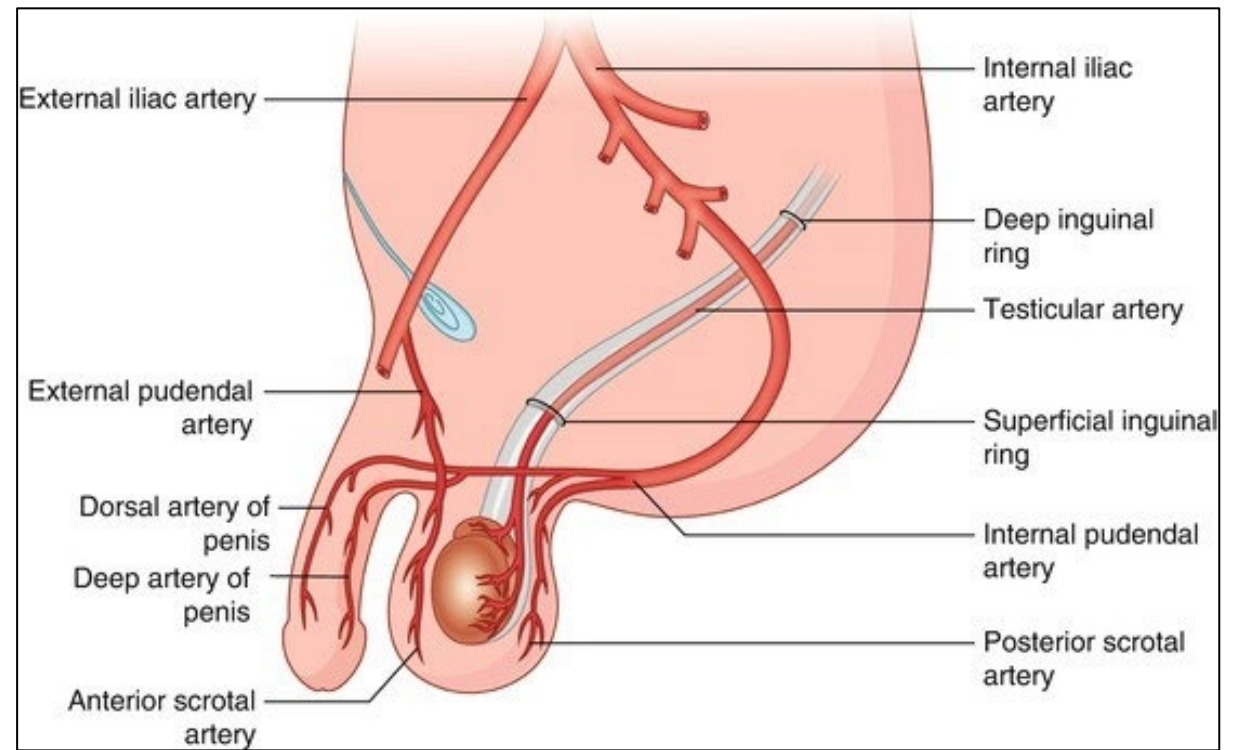
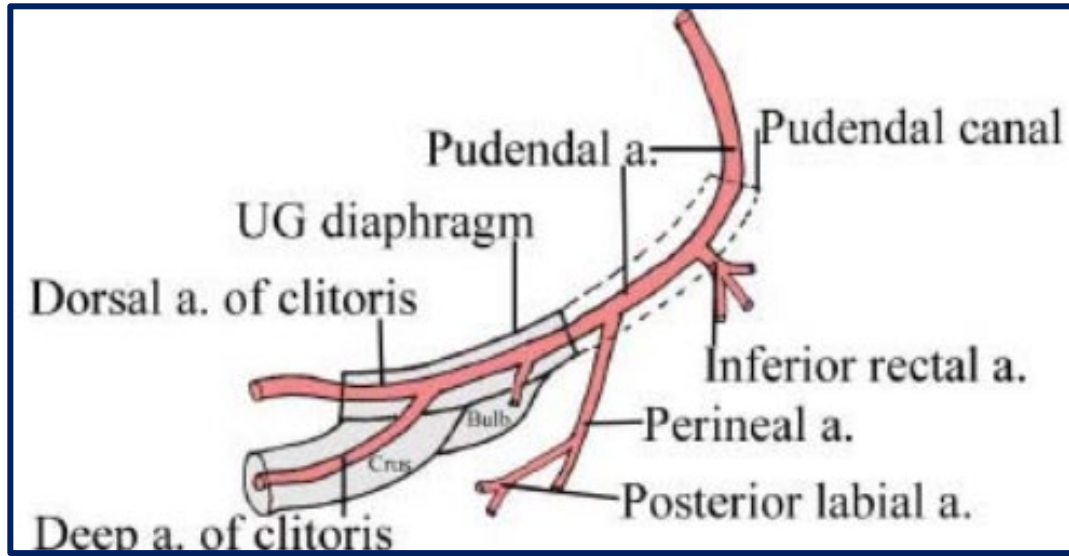




Nerves of the Perineum



Arteries of the Perineum

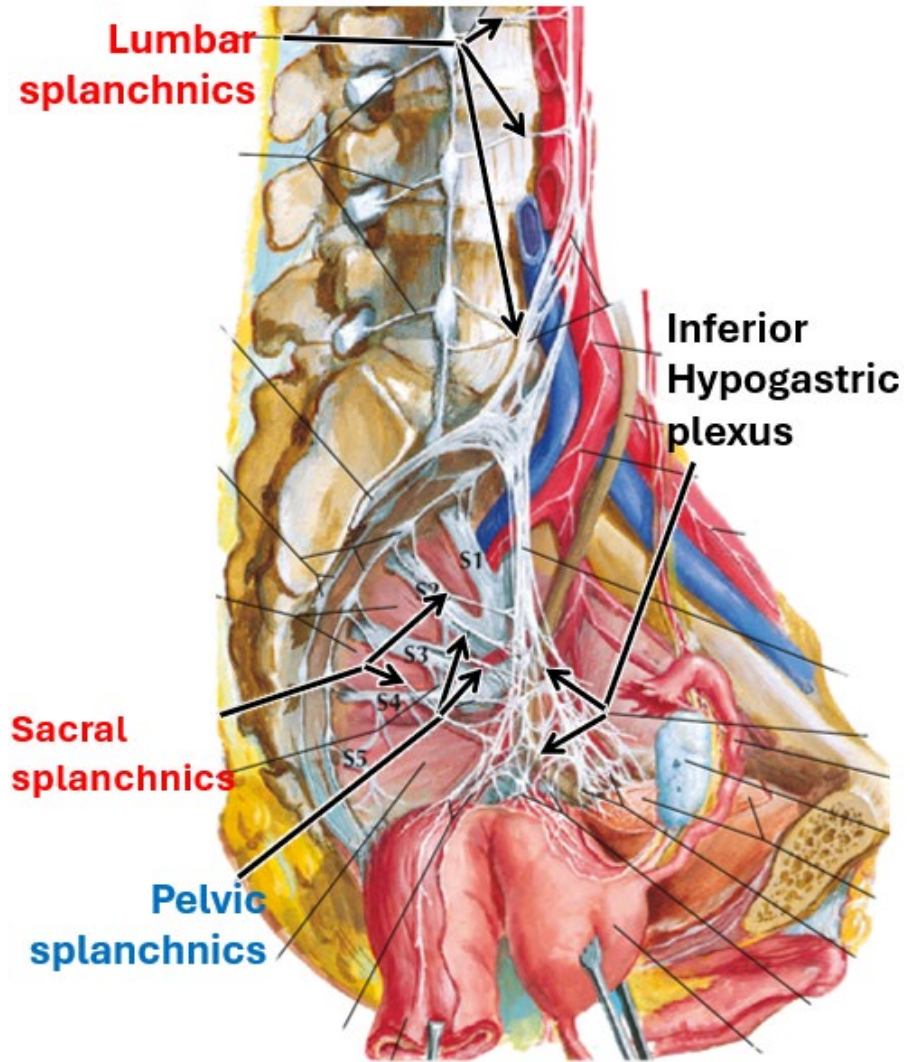


- **Inferior rectal a.** to the anal canal inferior to the pectinate line, anal sphincter & peri-anal skin
- **Perineal a.** to superficial perineal muscles

- **Posterior labial a.** to skin of posterior labia
- **A. of bulb of vestibule** to bulb of vestibular bulbs and greater vestibular glands
- **Deep a of clitoris** to crura of clitoris
- **Dorsal a of clitoris** to glans of clitoris

- **Posterior scrotal a** to skin of posterior scrotum
- **A. of bulb of penis** to supply bulb, penile urethra, corpus spongiosum, bulbourethral glands
- **deep a of penis** - travels through the center of each corpus cavernosum; gives off straight and helicine aa, which open directly into the sinusoids of the corpora.
- **dorsal a of penis** to corpora
- **superficial external pudendal a** to anterior scrotum

Innervation of Pelvis & Perineum



Sensory & motor innervation of skin, skeletal muscle

- Somatic division of PNS: **sacral plexus** (L4-S4)
 - Nerve to obturator internus (L5-S1)
 - Nerve to piriformis (S1-S2)
 - Nerve to levator ani (S4, S5)
 - Nerve to coccygeus (S4, S5)
 - **Pudendal nerve (S2-S4)** (perineum)

Motor innervation of viscera

- Autonomic division of PNS: **inferior hypogastric plexus**
 - **Sympathetic innervation**
 - **Lumbar & sacral splanchnics:** effects – vasoconstriction, **contraction** of internal urethral (male) & anal sphincters, **stimulate** glandular secretion, ejaculation (male)
 - **Parasympathetic innervation**
 - **Pelvic splanchnics (S2-S4):** effects – vasodilation, contraction of the detrusor, engorgement of erectile tissue.

Autonomic Regulation of Sexual Function

Erection: parasympathetic (S2-4 via pelvic splanchnic nn)

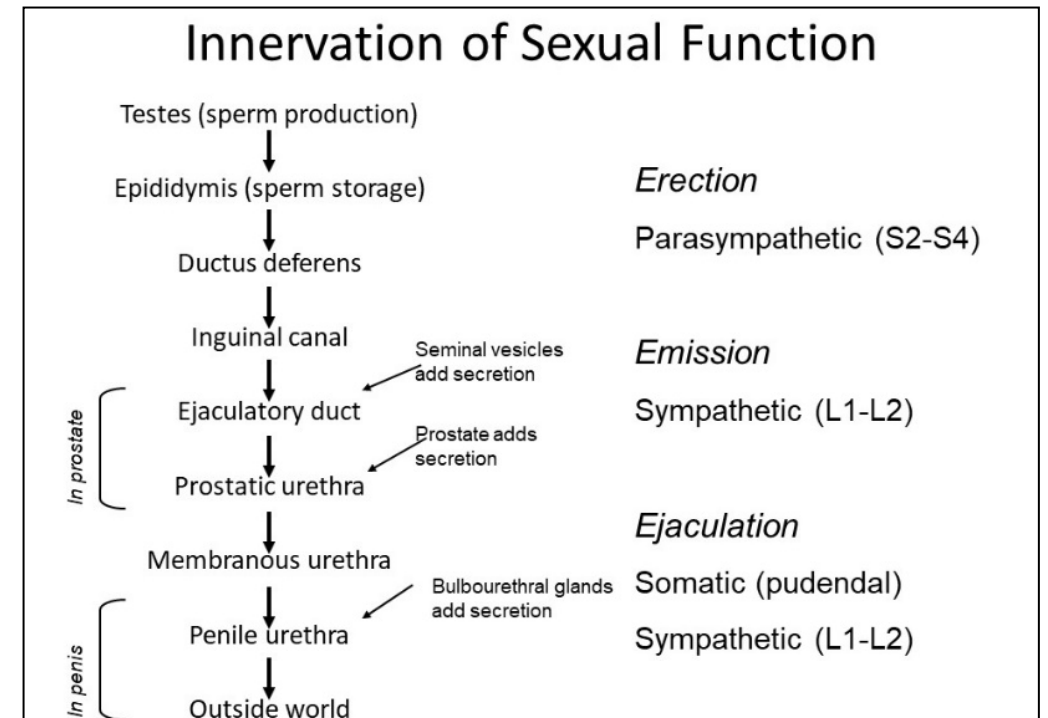
•**Emission:** sympathetic (L1-2)

Smooth muscle contraction of the ductus deferens to move sperm into the urethra. Contractions of seminal vesicles, prostate, and bulbourethral glands secrete seminal fluid into the urethra.

•**Ejaculation:**

•**Sympathetic (L1-2):** rhythmic contraction of smooth muscle in the ductus deferens & urethra; closure of internal urethral sphincter. **Somatic (S2-4, pudendal n):** contraction of bulbospongiosus & ischiocavernosus muscles compress the bases of the penile erectile tissue.

Point = Parasympathetic = Erection
Shoot = Sympathetic = Ejaculation



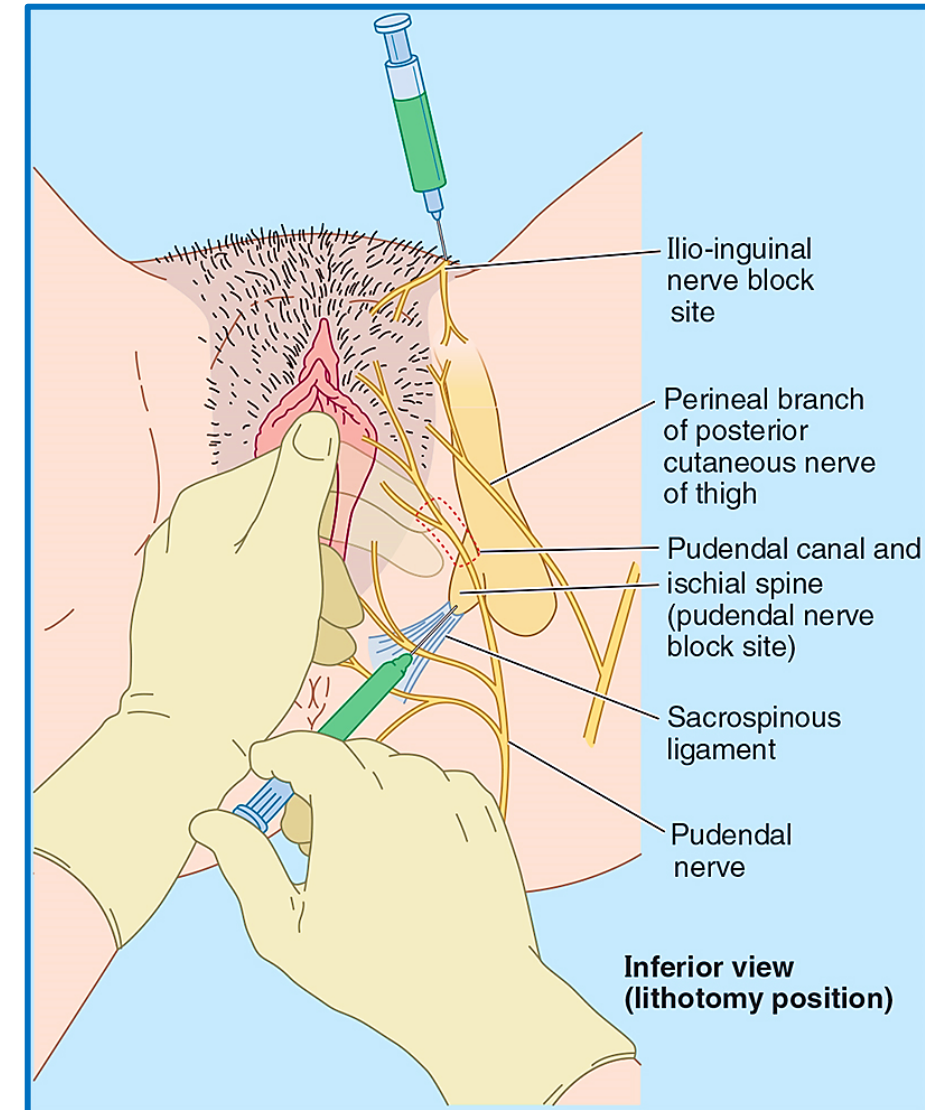
Pudendal Nerve Block

Pudendal nerve:

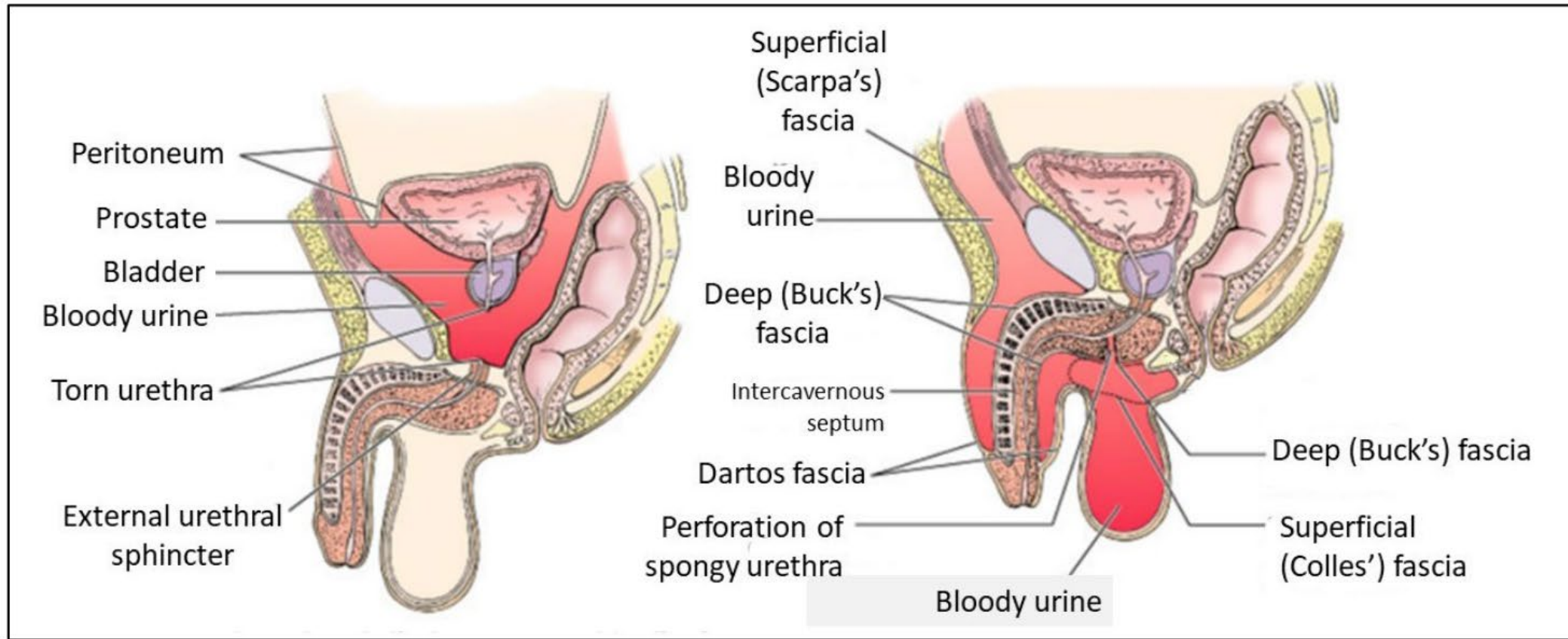
Passes out of the pelvis via the greater sciatic foramen & around the sacrospinous ligament into the lesser sciatic foramen, where it enters the pudendal (Alcock's) canal; provides somatic innervation (S2-S4) to the skin & skeletal muscles of the perineum & gives rise to inferior rectal (anal), perineal, labial, & dorsal clitoral nerve branches.

To relieve perineal pain during childbirth, ***pudendal nerve block anesthesia*** may be performed by injecting a local anesthetic agent into the tissues surrounding the pudendal nerve.

To abolish sensation from the anterior part of the perineum, an ***ilioinguinal nerve block*** is performed.



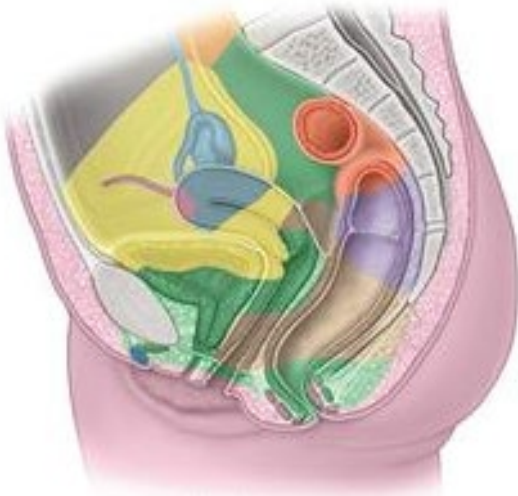
Male Urethral Rupture



Straddle injury: rupture of the spongy urethra below the perineal membrane. Urine extravasates to the scrotum, penis, superficial perineum, and lower anterior abdominal wall.

Pelvic fractures: rupture of the membranous (intermediate part) of the urethra. Urine extravasates to the deep perineal pouch.

Female Lymphatic Drainage

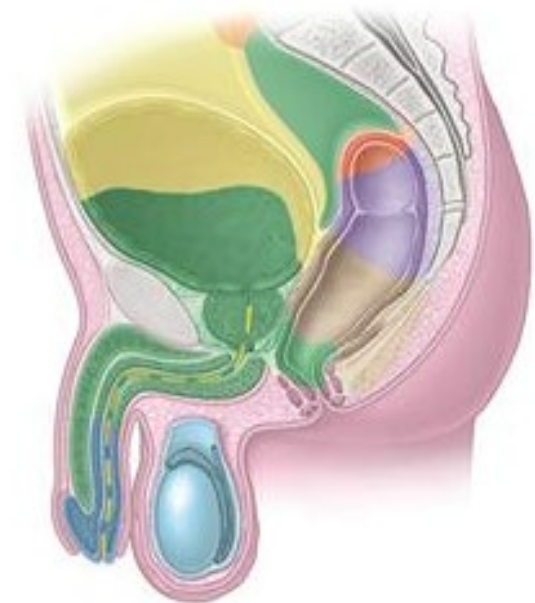


(B) Median section

Groups of lymph nodes	
Lumbar (caval/aortic)	Superficial inguinal
Inferior mesenteric	Deep inguinal
Common iliac	Sacral
Internal iliac	Pararectal
External iliac	Direction of flow

Lymph Node Group	Typically Drains
Lumbar (along ovarian vessels)	Gonads and associated structures, common iliac nodes (ovary, uterine tube except isthmus and intrauterine parts, fundus of uterus)
Inferior mesenteric	Superiormost rectum, sigmoid colon, descending colon, pararectal nodes
Internal iliac	Inferior pelvic structures, deep perineal structures, sacral nodes (base of bladder, inferior pelvic ureter, anal canal above pectinate line, inferior rectum, middle and upper vagina, cervix, body of uterus)
External iliac	Anterosuperior pelvic structures, deep inguinal nodes (superior bladder, superior pelvic ureter, upper vagina, cervix, lower body of uterus)
Superficial inguinal	Lower limb; superficial drainage of inferolateral quadrant of trunk, including anterior abdominal wall inferior to umbilicus, gluteal region, superficial perineal structures (superolateral uterus near attachment of round ligament, skin of perineum including vulva, ostium of vagina inferior to hymen, prepuce of clitoris, perianal skin, anal canal inferior to pectinate line)
Deep inguinal	Glans clitoris, superficial inguinal nodes
Sacral	Posteroinferior pelvic structures, inferior rectum, inferior vagina
Pararectal	Superior rectum

Male Lymphatic Drainage



(B) Median section

Groups of lymph nodes

Lumbar (caval/aortic)

Inferior mesenteric

Common iliac

Internal iliac

External iliac

Superficial inguinal

Deep inguinal

Sacral

Pararectal

Direction of flow

Lymph Node Group	Typically Drains
Lumbar (near testicular vessels)	Urethra, testis, epididymis
Inferior mesenteric	Superiormost rectum, sigmoid colon, descending colon, pararectal nodes
Internal iliac	External and internal iliac lymph nodes
External iliac	Inferior pelvic structures, deep perineal structures, sacral nodes (prostatic urethra, prostate, base of bladder, inferior pelvic ureter, inferior seminal glands, cavernous bodies, anal canal above pectinate line, inferior rectum)
Superficial inguinal	Lower limb; superficial drainage of inferolateral quadrant of trunk, including anterior abdominal wall inferior to umbilicus, gluteal region, superficial perineal structures (skin of perineum, including skin and prepuce of penis, scrotum, perianal skin, anal canal inferior to pectinate line)
Deep inguinal	Glans penis, superficial inguinal nodes, distal spongy urethra
Sacral	Posteroinferior pelvic structures, inferior rectum
Pararectal	Superior rectum



Questions

Question

A 55-year-old man complains of severe pain at his anus. Examination reveals thrombosed external hemorrhoids below the pectinate line. The physician explains that the pain is transmitted by somatic sensory fibers supplying the anal canal. Which of the following nerves carries pain sensation from this patient's hemorrhoids?

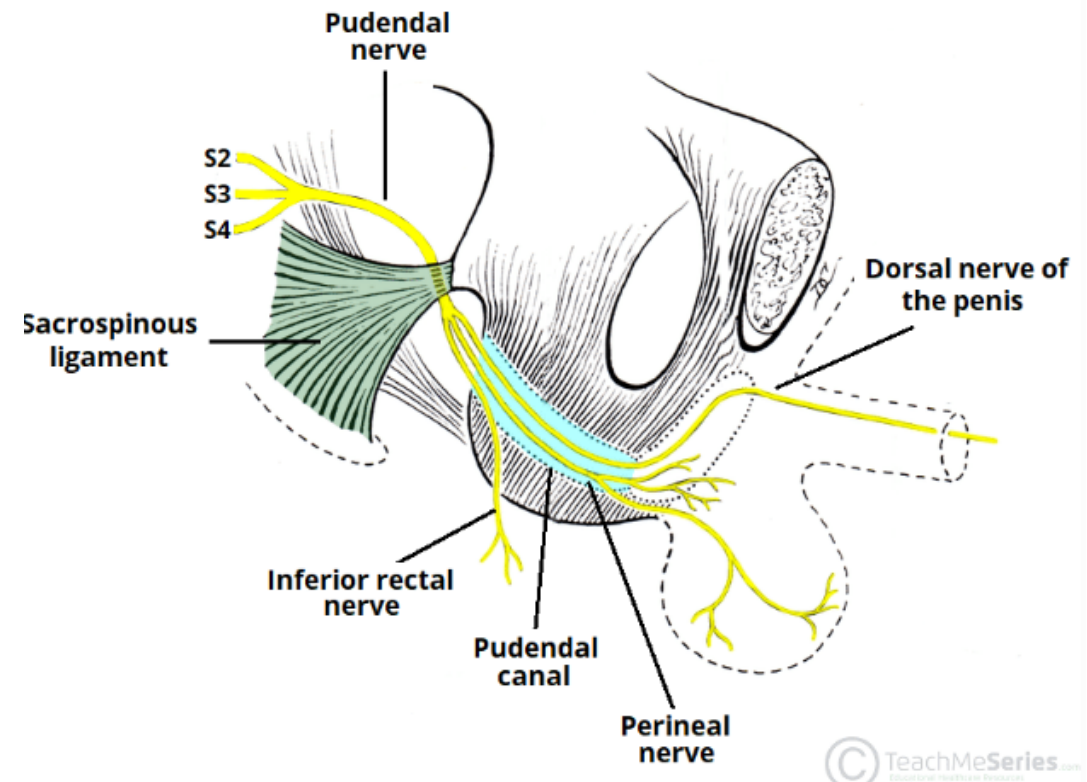
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- B. Sacral splanchnics
- C. Perineal nerve
- D. Inferior rectal nerve
- E. Posterior scrotal nerve

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D



Question

A 14-year-old boy is admitted to the emergency department 2 days after crashing his bicycle. MRI examination reveals severe edema of the boy's scrotum and abdominal wall and extravasated urine. Which of the following structures is most likely ruptured?

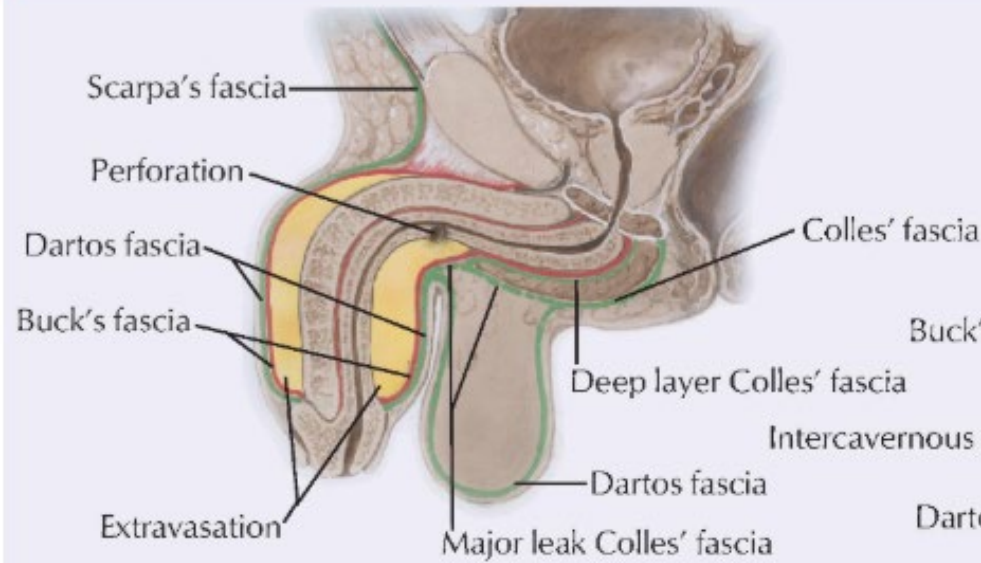
- A. Spongy urethra
- B. Preprostatic urethra
- C. Prostatic urethra
- D. Urinary bladder
- E. Ureter

Question

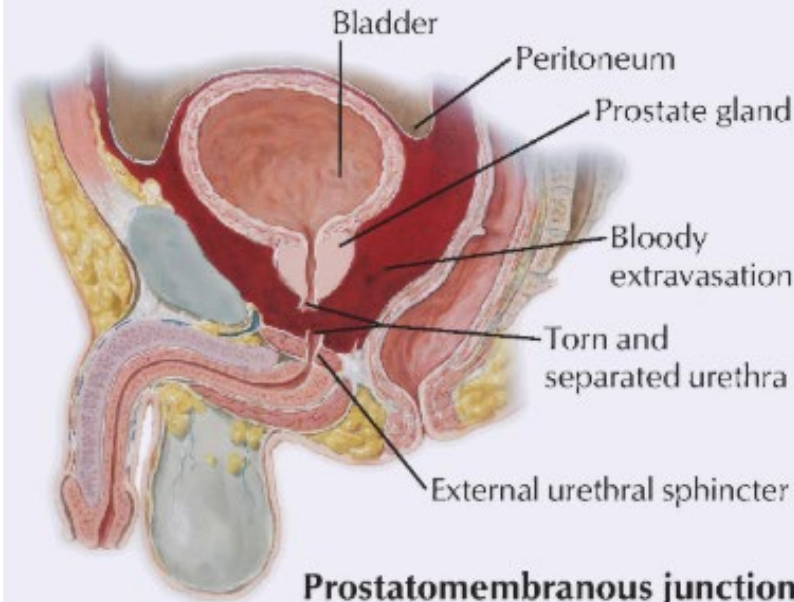
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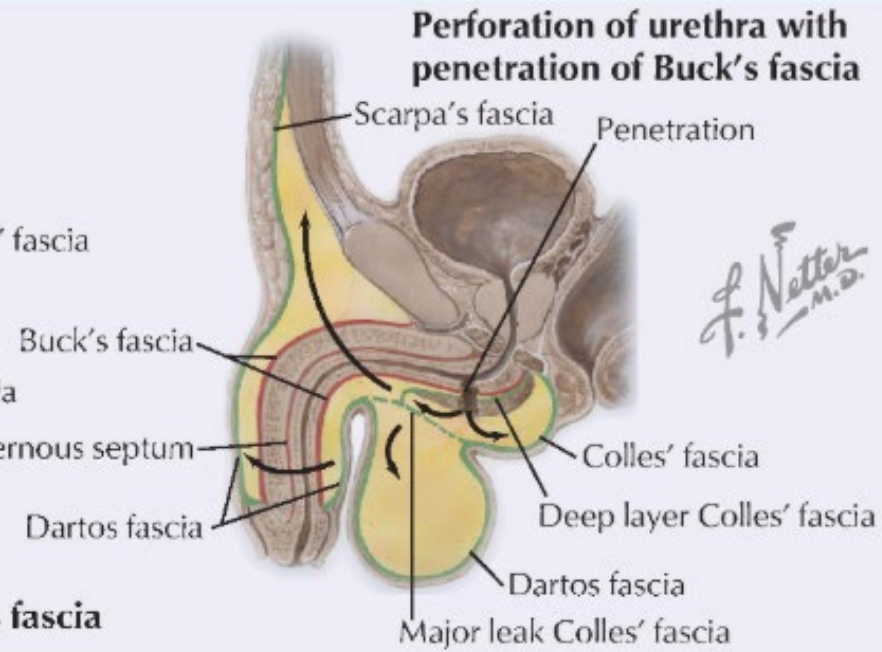
A



Perforation of urethra without penetration of Buck's fascia



Prostatomembranous junction injury



Injury site	Consequences
Penile urethra, Buck's fascia intact	Localized swelling confined to penis
Penile urethra, Buck's fascia ruptured	Eventual collection of urine deep to Colles' fascia (superficial perineal fascia); perineum: superficial pouch; penis; scrotum: deep to dartos (superficial) fascia; lower abdominal wall: deep to Scarpa's fascia
Prostatomembranous junction	Potential injury with anterior pelvic fractures, which may lead to retroperitoneal hematoma and urine extravasation

Question

A 34-year-old man is admitted to the emergency department after a traumatic landing into a swimming pool from a high diving platform. The patient has multiple traumas in his abdominal cavity. After a reconstructive operation, the patient develops a high fever. Radiographic examination reveals that the lower portion of the descending colon and rectum has become septic and must be excised. Six months postoperatively, the patient complains of impotence.

Which of the following structures was most likely injured during the second operation?

- A. Pudendal nerve
- B. Sacral splanchnic nerves
- C. Pelvic splanchnic nerves
- D. Sympathetic trunk
- E. Vagus nerve

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C

Point = Parasympathetic = Erection
Shoot = Sympathetic = Ejaculation

Question

A 34-year-old woman was admitted to the labor and delivery ward during the second stage of labor. The obstetrician noted that the descent of the fetus was arrested as the head seemed not to be able to pass the vaginal orifice. The decision was taken to perform a surgical incision of the perineum to enlarge the vaginal orifice, thereby decreasing the risk of tearing of the perineum and its muscles. An oblique incision was made to the right of the midline. Which structure is most likely maintained?

- A. External anal sphincter
- B. External urethral sphincter
- C. Right superficial transverse perineal muscle
- D. Right ischiocavernosus muscle
- E. Perineal body

Question

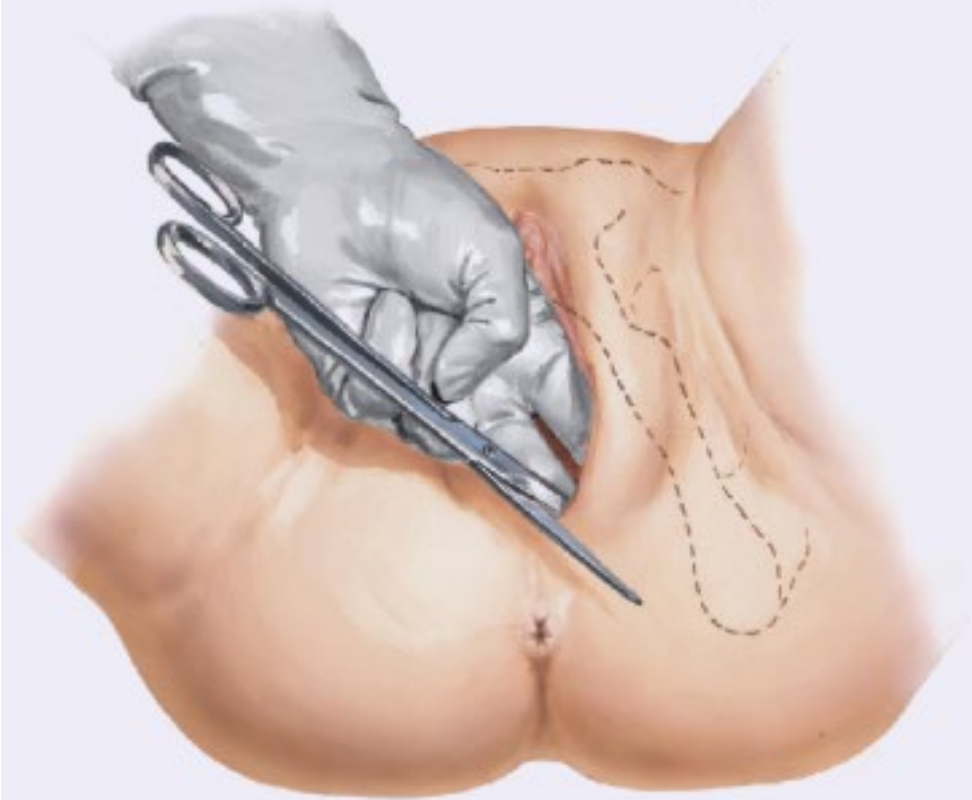
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E

Posterolateral Approach

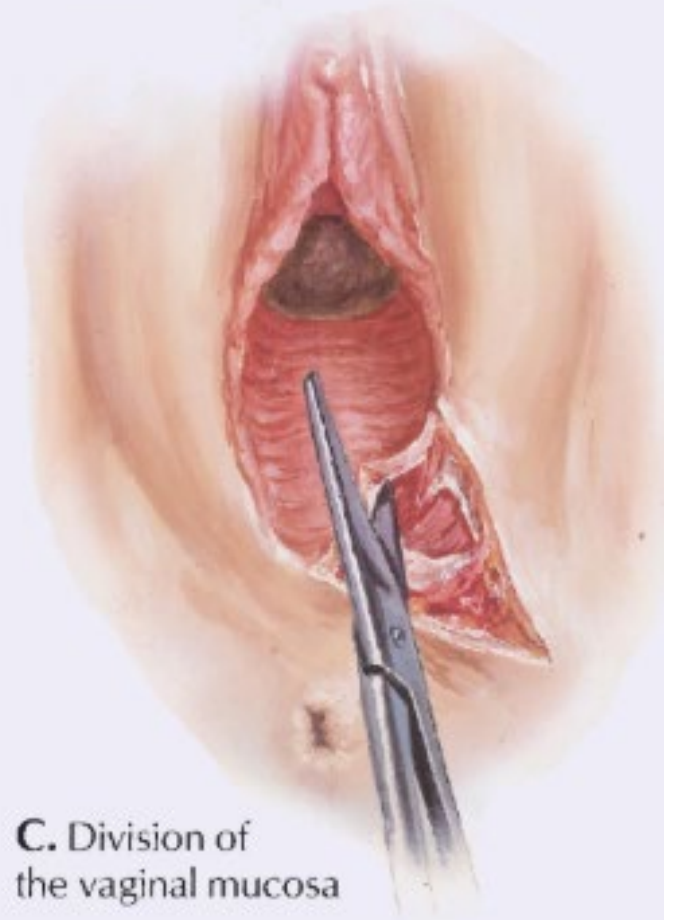
(one of two approaches normally used, posterolateral [shown here] or median)



A. Scissors are directed from the midline to the tuberosity



B. Incision in the bulbospongiosus muscle



C. Division of the vaginal mucosa

Question

A 66-year-old woman with multiple sclerosis suffers from urinary incontinence. Neurological examination demonstrates loss of voluntary control of the external urethral sphincter.

Which of the following branches is most likely the cause of her incontinence?

- A. Inferior rectal nerve
- B. Deep branch of the perineal nerve
- C. Posterior labial nerve
- D. Superficial perineal nerve
- E. Dorsal nerve of the clitoris



Question

A 66-year-old woman with multiple sclerosis suffers from urinary incontinence. Neurological examination demonstrates loss of voluntary control of the external urethral sphincter.

Which of the following branches is most likely the cause of her incontinence?

B

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- B. Deep branch of the perineal nerve
- C. Posterior labial nerve
- D. Superficial perineal nerve
- E. Dorsal nerve of the clitoris

Pudendal Nerve (S2-S4) Branches

Inferior rectal nerve → External anal sphincter, anal canal below the pectinate line

Perineal nerve (deep branch) → External urethral sphincter and perineal muscles

Perineal nerve (superficial branch) → Posterior scrotal/labial sensory innervation

Dorsal nerve of penis/clitoris → Sensory innervation of erectile tissues

Question

A 34-year-old woman presented with progressive lower abdominal pain & abdominal distention. Radiographic examination reveals a malignant ovarian tumor. Which of the following lymph nodes will most likely become invaded by cancerous cells?

- A. Superficial inguinal
- B. External iliac
- C. Lumbar/aortic
- D. Deep inguinal
- E. Internal iliac

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A 34-year-old man presents with a painful erythematous swelling on the posterior aspect of the scrotum. Physical examination reveals a fluctuant abscess involving the scrotal skin. On further evaluation, enlargement of the regional lymph nodes draining the affected area is noted.

Which of the following lymph node groups is most likely to be tender and enlarged in this patient?

- A. External iliac lymph nodes
- B. Internal iliac lymph nodes
- C. Lateral aortic (lumbar) lymph nodes
- D. Sacral lymph nodes
- E. Superficial inguinal lymph nodes

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E

Structure	Lymphatic Drainage
Ovary	Para-aortic (lumbar) nodes
Testis	Para-aortic (lumbar) nodes
Scrotum	Superficial inguinal nodes
Labia majora	Superficial inguinal nodes
Glans penis/clitoris	Deep inguinal nodes

The background of the image is a close-up of an ancient Egyptian stone wall. It features several rows of hieroglyphs carved into the light-colored, textured stone. The hieroglyphs are arranged in vertical columns. Some of the visible symbols include a lotus flower, a scepter, a was staff, a djed staff, a falcon, and a seated figure. The lighting is warm, highlighting the relief of the carvings.

THANK YOU